

UNIVERSIDADE DE LISBOA
FACULDADE DE MEDICINA VETERINÁRIA



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APPLICATION OF SYNDROMIC SURVEILLANCE TO ROUTINELY COLLECTED SWINE
PRODUCTION DATA FOR FARM HEALTH MANAGEMENT AND EARLY DISEASE
DETECTION

CAROLINA MENDES GALANTE MERCA

ORIENTADORA:
Doutora Fernanda Dórea

COORIENTADORA:
Doutora Yolanda Maria Vaz

2021

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CAROLINA MENDES GALANTE MERCA

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Aluno: Carolina Merca

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DN: C=SE, OU=SVA, O=SVA,
CN=Fernanda Dórea,
E=fernanda.dorea@sua.se
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Dedicado à minha mãe, feito com o trabalho,
brio e amor com que sempre me ensinaste.
Sei que estarás orgulhosa.

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I give you all my most sincere gratitude.

RESUMO

Aplicação de vigilância sindrómica em dados de produção de suínos recolhidos de forma rotineira para manejo da saúde da exploração e para deteção precoce de doenças

No contexto atual de risco acrescido de aparecimento de novas doenças infecciosas que acarretam um impacto significativo na saúde pública humana, saúde animal e economia global, a vigilância sindrómica (VS) surge como uma abordagem promissora para a deteção precoce de episódios de doença inesperados.

O presente projeto consiste na construção de um sistema de VS com o objetivo de fornecer informação relevante no âmbito do manejo da produção e da saúde animais. Para isso, foram utilizados dados de produção de suínos recolhidos de forma rotineira através do software WinPig, e compartilhados voluntariamente por uma exploração sueca de reprodução. A metodologia adotada reside na aplicação combinada dos gráficos de controlo EWMA e Shewhart em vários indicadores relacionados com a saúde animal, por forma a detetar acontecimentos temporais considerados anormais no desempenho reprodutivo. Foram utilizados três limites de deteção baseados na pontuação de severidade do alerta. Adicionalmente, foi construída uma plataforma com os resultados do sistema de VS para utilização futura por produtores e médicos veterinários, para que estes possam perceber se a exploração em estudo se encontra a gerar alarmes, permitindo-lhes atuar atempadamente.

A eficácia do sistema de VS na deteção precoce de doenças foi avaliada utilizando dados simulados com injeção de surtos de Síndrome reprodutivo e respiratório suíno (PRRS). Concluiu-se que os surtos injetados tinham elevada probabilidade de serem notados pelos utilizadores do sistema e numa fase precoce da doença. Embora estes resultados não devam ser extrapolados de forma acrítica para outras doenças e explorações, realçam a utilidade da VS e da utilização dos dados de produção para o alerta precoce de doenças, tendo um grande potencial para desenvolvimentos futuros no campo da vigilância da saúde animal.

Palavras-chave: Vigilância sindrómica, dados de produção, gráficos de controlo, deteção de aberrações temporais, deteção precoce de doenças.

ABSTRACT

Application of syndromic surveillance to routinely collected swine production data for farm health management and early disease detection

Within the current context of increased risk for the emerging of new infectious diseases, with a great impact on both human public health and animal health and global economy, syndromic surveillance (SyS) appears an increasingly more suitable approach for the early detection of unexpected health events.

This project embraces the construction of a SyS system giving relevant information for productivity and health management. Swine production data routinely collected by WinPig software and voluntarily shared by a Swedish reproduction farm, was the data source used. The chosen methodology employed was the combination of EWMA and Shewhart control charts on several health-related indicators to detect temporal aberrations of reproductive performance, using three detection limits based on a “severity” score. Furthermore, an interface with the SyS results was constructed for future use by farmers and veterinarians, where they can notice if the farm under study is experiencing alarms, allowing them to act timely.

The SyS system’s effectiveness in early detection of diseases was evaluated using simulated data with the injection of Porcine respiratory and reproduction syndrome (PRRS) outbreaks. It was concluded that the injected outbreaks had a high chance of being noticed and at an early phase of the disease. Although these results should not be incautiously extrapolated to other diseases and farms, it corroborates the usefulness of SyS and production data to early disease warning with great potential for further development in the field of animal health surveillance.

Keywords: Syndromic surveillance, production data, control charts, temporal aberration detection, early disease detection.

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List of acronyms, abbreviations and symbols

BTV-8 – Bluetongue virus serotype-8

CC – Control charts

CUSUM – Cumulative Sums

DGAV – Portuguese General Directorate of Food and Veterinary (Direção-Geral de Alimentação e Veterinária)

DGS – Portuguese General Directorate of Health (Direção-Geral da Saúde)

ESCAIDE – European Scientific Conference on Applied Infectious Disease Epidemiology

EWMA – Exponentially Weighted Moving Average

FMV-ULisboa – Faculty of Veterinary Medicine of the University of Lisbon (Faculdade de Medicina Veterinária da Universidade de Lisboa)

G&D – Farm and Animal Health (“Gård och Djurhälsan”)

LCL – Lower control limit

PRRS – Porcine respiratory and reproduction syndrome

SBA – Swedish Board of Agriculture

SBV – Schmallenberg virus

s.d. – Standard deviation

SISS – Swine Health Information System (Sistema Informativo da Sanidade dos Suínos)

SVA – Swedish National Veterinary Institute (Statens veterinärmedicinska anstalt)

SyS – Syndromic Surveillance

TADA – Temporal aberration detection algorithms

UCL – Upper control limit

VSS – Veterinary syndromic surveillance

± – Plus or minus

≥ – Greater than or equal to

≤ – Less than or equal to

1. Internship Report

The curricular internship was performed at the National Veterinary Institute (Statens veterinärmedicinska anstalt, SVA) in Uppsala, Sweden, from January 18th to May 5th, 2021, with Erasmus+ Program's financial aid. This training was accomplished under the Department of Disease Control and Epidemiology, supervised by Doctor Fernanda Dórea and co-supervised by Professor Yolanda Vaz, with a total amount of 620 hours. This dissertation with the title: "Application of syndromic surveillance to routinely collected swine production data for farm health management and early disease detection" was part of a research study, PigPeaks Project, conducted at the host institution.

Due to the COVID-19 pandemic, I was unable to be physically present at the SVA facilities daily, but it was possible to go to SVA a few times to meet the team and to attend some meetings in person with my supervisor. Regardless of this situation, I was able to develop different skills such as literature review, coding and data analysis using the statistical programming language R, understanding the algorithms, how they operate and their applicability, and develop my oral presentation capabilities with open discussion of the present results. I also learnt how to work with R packages, {vetsyn} to create automatized processes of converting animal health data into epidemiological information, and {flexdashboard} to build dashboards. Some errors were found in the {vetsyn} package and they were therefore corrected after studying the codes which are at its origin. In parallel, I was able to meet SVA's swine specialist Doctor Marie Sjölund from the Department of Animal Health and Antimicrobial Strategies, who provided valuable input regarding the project, contributing with her own perspective and personal experience. Additionally, since the beginning of March 2021, an online Epidemiology Journal Club was created and attended by myself and both former and current faculty's colleagues also enthusiastic about the epidemiology field. It has been taking place on Thursdays every 2 weeks, where we discuss and comment published scientific articles in order to keep up with the current literature, promote awareness of current research findings and train the science of debate.

From September 7th to December 18th, 2020, to reinforce the training in the veterinary epidemiology field before the curricular internship at SVA, some competencies regarding the use of R programming and MySQL Database Service, using SQL language, were developed at the Faculty of Veterinary Medicine of the University of Lisbon (FMV-ULisboa), under Professor Telmo Nunes' supervision. As the world has been experiencing an unprecedented pandemic, it was decided to analyze the COVID-19 data in Portugal to gain competencies of data analysis using the statistical programming language R. Then, challenged by Professor Telmo Nunes, a project called "Mobility's Influence on Increasing the Number of New Cases of COVID-19 in Portugal" was performed with colleague Raquel Costa's collaboration and was

sent to the Director of Information and Analysis Services of the Portuguese General Directorate of Health (DGS) who was impressed with the work done. Afterward, the database of Swine Health Information System (SISS) from the Portuguese General Directorate of Food and Veterinary (DGAV) was used as training for using MySQL to extract and cluster data from the relational database. Also, an online course by the name of "Introduction to statistics & data analysis in R for public health" by Imperial College London available in the Coursera platform was completed. During this period, I had weekly participation in webinars with former FMV-ULisboa's students about their academic and professional backgrounds which contributed to broaden my horizons and to enrich my personal understanding of this profession's possibilities. I also attended the European Scientific Conference on Applied Infectious Disease Epidemiology (ESCAIDE), which took place online from November 24th to November 27th, 2020.

Additionally, during these 4 months, it was possible to perform an extracurricular traineeship in cardiology, once or twice a week, at the University Teaching Hospital, FMV-ULisboa where I gave support to cardiology's appointments of small animals, including echocardiography's execution.

2. Introduction

An increase in incidence of infectious diseases has been experienced worldwide in both human and veterinary medicines. This is possibly due to the present context of globalization, global warming, animal and human population growth, bioterrorism threat, among others. These diseases represent a significant burden on global economies and public and animal health (Jones et al. 2008), creating a need for investment in new approaches of surveillance programs that can detect the emergence of diseases as quickly as possible.

As a response to the need for timely disease detection, and enabled by the growing data digitalization, syndromic surveillance (SyS) has been used more and more frequently in public health (Mandl, Overhage, et al. 2004) and animal health (Dórea et al. 2011) since the beginning of the millennium. The Triple-S project (Syndromic Surveillance Systems in Europe), co-financed by the European Commission, defined SyS as “a real-time (or near real-time) collection, analysis, interpretation, and dissemination of health-related data to enable the early identification of the impact (or absence of impact) of potential human or veterinary public health threats that require effective public health action” (Triple-S project 2010)¹.

This project aimed to investigate whether production data recorded regularly in reproduction pig farms could be used to construct health and performance indicators; and whether these indicators could be subjected to near-real time trend monitoring (syndromic surveillance) to detect early signals of reproduction failure, which could be caused by undetected diseases or management errors.

In chapter 3, a literature review will be presented, addressing the syndromic surveillance, the development of such system and its use, and the swine sector in Sweden and information systems used for farm management. Chapter 4 and following chapters, present the work carried out with the specific objectives, the materials and methods, results, discussion and conclusions.

¹ Triple-S. Triple-S—the syndromic surveillance project. <http://www.syndromicsurveillance.eu>

3. Literature Review

3.1. Syndromic surveillance

The emergence of new pathogens and the growing threat of bioterrorism have motivated the development, especially since the late 1990s, of surveillance systems that can provide timely detection. Initial work in the field focused on identifying data that could have signs of disease introduction. In that regard, pre-diagnostic data (data that precede diagnostic confirmation) seemed to be usable since it is early available. These data can be aggregated into syndromes; an approach that led to an increase in the use of the term “syndromic surveillance” (SyS) (Shmuel and Burkom 2010; Dórea, McEwen, McNab, Sanchez, et al. 2013). The prime statement of any SyS system is that a given indicator of population health changes when a health hazard is introduced, and these changes can be noticed if routinely collected data are analyzed (Dórea et al. 2011).

3.1.1. Steps to construct a syndromic surveillance system

3.1.1.1. Data source

Adequate data for SyS should be routinely collected, in an automated manner, electronically stored and timely accessible to ensure the success of the program (Mandl, Overhage, et al. 2004). The Triple-S project revealed that, in fact, there is an increase in the collection of data suitable for veterinary SyS in Europe (Triple-S project 2010)¹. As electronic records are reaching farms, data recorded at the herd level, or even individually for each animal regarding different phases of productivity (hereby generalized as “production data”), have appeared as a growing data source for animal health surveillance. These data, if correctly managed, allow large population coverage and the shortest time lag between a health event and its potential detection. Other data sources used in SyS are laboratory data, abattoir, media sources, among others (Gates et al. 2015; Dórea and Vial 2016).

3.1.1.2. Create time-series

When adequate data for SyS are available, the next step is to choose the observations to monitor and aggregate them within a set time window (daily, weekly, or monthly) to construct a time-series. A time-series is an ordered sequence of observations. In most cases, it is data which is taken only at specific time intervals equally spaced in time (discrete time-series). In opposition, observations that can be recorded continuously are called continuous time-series (Wei 1991). Most of the SyS systems developed to date, in particular in animal health, are based on discrete time-series (Dórea and Vial 2016) and the literature reviews of methods for

SyS have focused on discrete time-series analysis (Unkel et al. 2012). The appropriate time unit to aggregate the data needs to be evaluated based on the data available. Shorter time units – such as daily – will result in more timely detection of aberrations, but implementation in practice is only possible if data are expected to be recorded and updated within the same day of the event. Moreover, statistical monitoring can be challenging if the number of events per day is too low. The choice should therefore be informed by the data collection procedures, the statistical properties of the data, and the desired timeliness of the system (Dórea, Revie, et al. 2013).

3.1.1.3. Retrospective analysis

Retrospective analysis is the step where available, historical data is used to learn as much as possible about the expected behaviour of the indicator. Understanding the known, repeatable patterns in the indicator time-series, such as seasonal effects and global trends, is a crucial step (Lotze et al. 2008). When they exist, these patterns should be modelled and removed (pre-processing) in order to separate the ordinary background behavior from temporal aberrations present in the data (Dórea, Revie, et al. 2013). Temporal patterns are generally modelled using regression models, most commonly generalized linear models, with the choice of family being based on the distribution of the data available (for example gaussian models for continuous data, poisson models for count data, and negative binomial models for zero inflated data) (Lotze et al. 2008).

3.1.1.4. Create a baseline

A baseline is needed to train the detection algorithms with the typical background behavior against which the presence of abnormalities is investigated. A decision has to be made regarding which part from the historical data is going to be used as baseline. A meaningful baseline can only be made when sufficient historical data is available. Shorter baselines give local effects too much weight, reducing sensitivity and allowing aberrations to contaminate the baseline. Longer baselines better capture global behaviour and smooth local noise, however the inclusion of events too far in the past may be a poor representation of the current behaviour of the indicator time-series (Dórea, McEwen, McNab, Revie, et al. 2013). No general rule exists to the length of the baseline, and the balance needs to be tailored based on the results of retrospective analysis.

Then, an outbreak-free baseline has to be created. Dórea and Vial 2016, reviewed methods used in animal SyS and showed that two main methods of baseline cleaning have been used. When the time-points of past outbreaks are known, regression models can be

applied with observations weighted by the inverse of their residuals, effectively reducing the contribution to the model of observations which largely deviate from the expected values as proposed by Farrington et al. 1996. When possible outbreaks are not labelled, the alternative option commonly used is to fit models to the entire period of data available, and eliminate outliers and excessive noise by substituting values above a certain confidence interval (generally 95%) by the cut-off value of that interval. For the latter option, both parametric (based on regression) and non-parametric (based on percentiles) methods are available (Dórea, Widgren, et al. 2015). A cut-off based on confidence limits rather than average behaviour is advised to keep as much of the original data as possible, and avoid over-smoothing of the data. Replacing values by the model prediction, rather than the model confidence limit, could generate a baseline that leads to the detection of an excessive number of false alarms (Dórea, Revie, et al. 2013)

To avoid contamination of the baseline with an early phase of an undetected outbreak (and therefore not yet corrected) it is suggested to leave a gap, or guard-band, between the baseline and the values to be evaluated (Burkom 2003; Murphy and Burkom 2008).

3.1.1.5. Apply temporal aberration detection algorithms (TADA)

Adequate algorithms must be chosen to distinguish if an observed event is significantly different from the expected levels defined by the cleaned historical baseline. When that difference is considered significant, an alarm is generated.

In the literature of veterinary syndromic surveillance (VSS), three main types of aberration detection algorithms have been explored: regression based methods, regression based methods with control charts, and control charts (CC) (Dórea and Vial 2016).

Regression based models, such as Holt-Winters, have been used extensively as prediction methods in SyS. When substantial historical data exists, these models can handle temporal effects directly without needing pre-processing (Elbert and Burkom 2009).

On the other hand, control charts are only appropriate for SyS if underlying temporal patterns are absent or removed. Based on documented literature in VSS, the most commonly used control charts in biosurveillance are: Exponentially Weighted Moving Average (EWMA), Cumulative Sums (CUSUM) and Shewhart charts. EWMA chart monitors a weighted average of the data with exponentially decaying weights according to its age and it is widely used in detecting gradual increases in the mean. The control charts CUSUM monitor cumulative sums of the deviations of the sample statistic from the target value. It is known to be efficient in detecting shifts in the process mean. Lastly, Shewhart chart evaluates a single observation and it is based on a simple calculation where a sample statistic is compared against the mean

or standard deviations. Shewhart is appropriate for detecting single spikes in the data (Yahav et al. 2011; Dórea, McEwen, McNab, Revie, et al. 2013).

EWMA deserve special mention for their ability to adjust rapidly to local trends without complex modelling. As it is an exponential algorithm, a smoothing coefficient called lambda is necessary. It determines the relative weight of current data to past data in order to define the rate at which older data influences EWMA predictions. It is recommended to use weights between 0.05 and 0.25, with 0.05, 0.1, and 0.2 being the most popular choices (Montgomery 2009).

A typical control chart, as illustrated in Figure 1, has a center line (CL) that represents the expectation value for the in-control process and two other horizontal lines, called the upper control limit (UCL) and the lower control limit (LCL). These control limits are chosen so that if the process is in control, the percentage of data points which will fall inside their boundaries represents the confidence level chosen (for instance 95% of data points within the LCL-UCL margins for a 95% confidence level). However, a point that falls outside of the control limits could be interpreted as evidence that the process is out of control, and investigation and corrective action are required to find and eliminate the assignable cause or causes responsible for this behavior (Montgomery 2009).

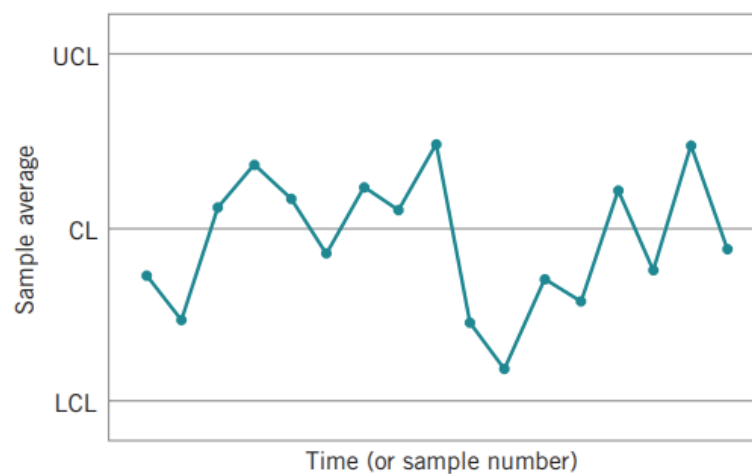


Figure 1: Typical control chart (Montgomery, 2009).

3.1.1.6. Parameterization and evaluation

In SyS control limits depend on the historical baseline of each individual farm. If a low threshold is set, the system may generate too many false alarms, and producers may stop paying attention to system alerts. If a high threshold is set, the system may not warn with enough sensitivity or timeliness when a health hazard has been introduced. Choosing the most suitable aberration detection algorithms for the data, and finding the best performance settings to achieve this balance between sensitivity and specificity is therefore a complex process, in particular since shapes of outbreak signals to be detected are unknown. Dórea, McEwen, McNab, Sanchez, et al. 2013 proposed a scoring system to combine multiple detection algorithms, each parameterized for multiple detection thresholds.

To assess the SyS system, ideally, real data should be used since the features of the baseline (such as temporal effects and noise) are expected to have a significant impact on system's performance. To evaluate the rate of false alarms, historical data without outbreak signals is needed, as addressed in the "create baseline" step described above. System sensitivity (true alarms), on the other hand, can only be evaluated when the TADA is challenged with data containing outbreak signals. This is challenging because historical data containing known outbreaks is not often accessible, or it is not possible to consistently identify the time-points associated with the outbreak in historical data. This point can be overcome by creating simulated data which includes the injection of artificial controlled outbreaks into the data. This strategy has the advantage of allowing the evaluation on outbreaks of different temporal distributions, sizes, and shapes (single spike, linear, exponential, log-normal) since it is uncertain the types of outbreaks likely to be faced in practice (Jackson et al. 2007; Dórea, McEwen, McNab, Revie, et al. 2013).

3.1.2. Syndromic surveillance as a complement to traditional surveillance

Traditional implemented surveillance systems for detecting emerging animal diseases are usually based on active or passive surveillance. Active surveillance consists of programs enforcing the regular testing of animals by governmental officials. These systems are effective in detecting the diseases they were designed for, but may not be able to detect the emergence of new threats (Dupuy et al. 2013; Dórea, Lindberg, et al. 2015). On the other hand, passive surveillance of animal diseases consists of suspicious of illness in one animal or in the herd, clinical diagnosis, and consequently report. These passive (clinical) surveillance systems have its focus on the detection of specific and previously described diseases and the advantage of providing coverage of the whole population at risk (RISKSUR project 2012). However, they may not be very effective in detecting diseases with non-specific clinical signs, diseases that

veterinarians or the animal technicians are not familiarized with, or the emergence of new diseases. The efficiency of the system is therefore determined by the awareness of the veterinarians or the animal technicians and their willingness to report the suspicion to the authorities (Veldhuis et al. 2016). The implementation of SyS systems that uses non-specific production data could serve as a “safety net”, and should be used “to reveal blind spots of traditional surveillance” (van den Wijngaard et al. 2011, p.6) instead of replacing the conventional surveillance systems already implemented. Moreover, they could provide a continuous view of the system performance (situational awareness). As such, SyS may be effective in detecting unspecific changes in herd health and could be applicable for early-warning disease surveillance (Hoinville et al. 2013; Veldhuis et al. 2016).

3.1.3. Existing syndromic surveillance systems

Dórea and Vial 2016 provided a review of the progress of animal health SyS systems. They showed that while few are implemented (beyond research), many different animal health data sources have been investigated such as clinical data, laboratory data, or media sources on companion animals, equine, cattle, and swine. These initiatives confirmed that there is a correlation between the outbreak occurrence and the observed variations in the syndromic indicators, validating the existence of an outbreak signal.

Production data, such as milk production and reproduction records, were also target of SyS research projects (Dórea and Vial 2016). More recently, Veldhuis et al. 2020 evaluated three statistical models for SyS using milk production data for the early detection of emerging diseases in cattle in the Netherlands. To assess the models, real outbreak data from the 2006 and 2007 epidemic Bluetongue virus serotype-8 (BTV-8) and the 2011 Schmallenberg virus (SBV) outbreak was used, since both vector-borne diseases have a negative effect on milk production. Particularly with the BTV-8 disease, alerts of drops in milk production were triggered long before first clinical suspicions were reported, remaining unknown if these alerts represent the actual first signs of BTV-8 infections. This shows that this data source type could be useful for early-warning, but as it is unspecific it could generate alarms for situations other than diseases.

Production data other than milk has not been extensively used in SyS, in particular for species other than cattle. Further investigation needs to be carried out since the use of production data could lead to earlier detection.

3.2. Swine production in Sweden

There are about 1150 pig producers registered in the Swedish Board of Agriculture (SBA), and 60% of these keep reproduction sows. Most of the total number of sows in this country (approximately 85%) are kept on just 26% of the farms, with an ongoing trend of farms becoming bigger (Pettersson 2021). Every year, 2.6 million pigs are slaughtered (The Swedish Board of Agriculture 2020).

Swine can be produced in different breeding systems. Sow herds can raise pigs from farrowing to slaughter, called farrow-to-finish farms, or producers can sell the pigs to specialized fattening farms. These farms do not house any sows and they are specific on fattening pigs (Pettersson 2021).

3.2.1. Production phases in reproduction pig farms

Swine production is generally classified into four production phases: Breeding/Gestation, Farrowing, Nursery, and Grow-finishing. Specialized reproduction pig farms only have the first three phases.

Regarding Breeding/Gestation, pregnancy length is usually three months, three weeks, and three days (114 days $\pm$ 3 days) and frequently the service is done by artificial insemination (Khalifa et al. 2014). A few days before farrowing, sows are moved to the farrowing units where they remain until their piglets are weaned (Pettersson 2021). The next production phase includes the farrowings and the weaning period. Swedish sows produce on average 2.2 litters of piglets per year. In each litter, an average of 14.6 piglets are born and out of those 12 are weaned (WinPig 2019)². After birth, piglets remain with the sow until weaning, which is not usually done before 28 days of age. In Sweden, the cross-fostering technique is used. It consists of fostering piglets from one sow to another within a farrowing group to increase piglet survival, and to create litters with more uniform body weight. The Nursery production phase, also called post-weaning, is the period from the time of weaning until around 12 weeks of age, and a weight of approximately 30 kg, being the piglets termed as weaners (Marie Sjölund, expert in pig farming in Sweden, personal communication).

² WinPig. 2019. WinPig Produktionsuppföljning.

3.2.2. Pig population health status

There has been a long-established strategy in Sweden to keep the country free from a number of infectious diseases and to strictly control the importation of live animals and semen (Carlsson et al. 2009). Swedish pigs are declared free from Aujeszky's disease (Robertsson 1996)³, Atrophic rhinitis (Wierup and Wallgren 2000)⁴, Porcine respiratory and reproduction syndrome (PRRS) (Carlsson et al. 2009), as well as from the diseases on the former list A of the World Organisation of Animal Health (OIE). However, in Europe, there has been a widespread occurrence of many diseases and an increase in trade, resulting in an impending threat of introduction of new pathogens in Sweden (Carlsson et al. 2009).

3.2.3. Digitalization of production data (WinPig)

The leading software for digital management of pig farms in Sweden is WinPig (known internationally as PigVision), produced by the Dutch company Agrosoft⁵. In Sweden, farmers using this software buy a license and yearly support from the advisory company "Gård och Djurhälsan (G&D)"⁶ (Farm and Animal Health), which keeps a database of safety backups for all companies. In Sweden, WinPig covers data from a few more than a third of the total sow population (Marie Sjölund, expert in pig farming in Sweden, personal communication).

Figure 2 shows a screenshot of the program main interface (in Swedish). In the program, every sow has a unique "sow card", where every life and reproduction event associated with that sow can be recorded, such as birth, inseminations, farrowing, death, etc. Events associated with litters of young piglets can also be recorded, but generally the litters are tracked in the software by the unique ID of the mother sow.

³ Robertsson J. 1996. Status of the national eradication program of Aujeszky's Disease. Swedish Veterinary Congress, Uppsala, Sweden. 179-184.

⁴ Wierup M, Wallgren P. 2000. Results of an intensive control of atrophic rhinitis in elite breeding and multiplier herds in Sweden. International Pig Veterinary Society Congress, Melbourne, Australia 16:158.

⁵ <https://www.agrosoftltd.com/pigvision/>

⁶ <https://www.gardochdjurhalsan.se/>

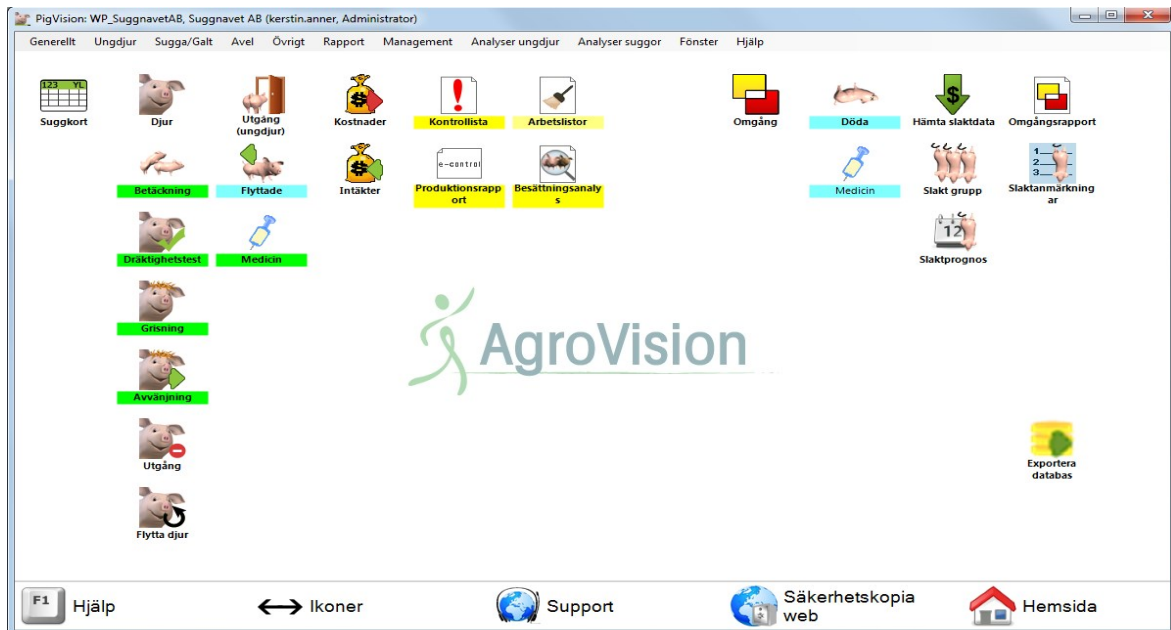


Figure 2: AgroVision's interface (in Swedish).

4. Objectives and justification of the work

The main objective of this project was to construct a syndromic surveillance system based on health and performance indicators, using production data recorded regularly in reproduction pig farms. Therefore, we aimed to detect early signals of reproduction failure, which could be caused by undetected diseases or management errors. The utility of such a monitoring system was further investigated by creating an interface to deliver results to target users. Target users can be farmers, directly, or their advisors, such as production experts and veterinarians.

By using data already regularly recorded in swine farming practice to extract and deliver, in a continuous and automated format, information relevant for productivity and health management, this project aimed to increase the value of these data for animal stakeholders. The detection of early signals of poor reproduction performance is expected to contribute to improve the cost-effectiveness of production. Moreover, the early detection of possible disease introductions can lead to improve animal welfare, public health as well as food safety.

5. Materials and methods

5.1. Access to the swine production data

This project was possible due to the collaboration of a large sow producing farm owner who voluntarily agreed to have their production records shared for this research. This farm uses the production management software WinPig, and their data were shared anonymously through the software vendor in Sweden (G&D). The researchers do not know the identity of the volunteering farm. The contributing farm focused on reproduction, and did not record information on finishing/fattening or slaughter. This project was therefore focused on reproduction events, as detailed below.

The data management structure of WinPig was investigated to explore possibilities to export the data. For back-end data management, WinPig creates a local server which hosts a SQL (Structured Query Language) database. Data from this SQL database were accessed within the statistical programming environment R (R Core Team 2020) using the package {RODBC} (Ripley and Lapsley 2020). Table 1 lists all the database tables accessed and the data extracted from them.

Table 1: Data available on the WinPig program and variables extracted from them.

Table name in the WinPig SQL database	Events recorded and structure	Variables extracted
animal	All life cycle events associated with each individual farm animal ^a	Animal unique identification in the system (Sow ID); date of birth; animal sex (to separate males and females); animal type (to identify active sows); date of exit; reason for exit
service	One row for each insemination event in the farm	Sow ID; date
abortion	One row for each abortion event recorded	Sow ID; date
farrowing	One row for each farrowing event recorded	Sow ID; date; number of piglets born (total, born alive, born dead, and when reported, number of mummified fetuses); piglets transferred to other sows or received from other sows (cross-fostering); when registered, average weight of piglets in the litter
weaning	One row for each weaning event recorded	Sow ID; date; number of piglets weaned; total litter weight; average weight per piglet
dead.piglet	One row for each day in which death in piglets was recorded	Sow ID; date; number of piglets dead
exit	One row for each sow leaving the farm	Sow ID; date; type of exit (from a number of specific categories, such as death, sold, slaughter); reason for exit (more detailed description of causes sometimes available, such as disease causes)

^a young piglets are not registered as individual animals, and litter information is associated directly with sows.

5.2. Data organization into indicators

The data available in WinPig were used to construct production and reproduction performance indicators. An initial list of all possible historical series that could be derived from the data presented in Table 1 was constructed, and this list was revised by stakeholders in a workshop that preceded the work reported in this document. Stakeholders selected a list of indicators that could be useful to monitor health, but were also interested in keeping some indicators valuable for production and reproduction management. Stakeholders also advised that the best unit of time for counting events and running detection would be *weekly*. Production advisors were of the opinion that daily monitoring would not be feasible because most producers do not enter information in WinPig daily, and weekly monitoring would provide a good balance between timeliness and visualization of long-term trends.

The developed monitoring system was therefore focused on these chosen indicators.

The indicators were divided into the following production phases: Services, Pregnancy, Farrowing, Weaning and Post-Weaning. Additional categories were created to group indicators related to sows before entering these production phases (Gilts), in between reproduction cycles (Empty Sows), and sows being removed from production (Exit). All available indicators, grouped by the listed categories, are presented in Table 2.

The first category, Gilts, has two indicators, both useful for farm management. For both of these indicators, *Age at first service* and *Age at first farrowing*, a new observation is recorded every time a sow experiences one of these events for the first time in their production life (insemination or farrowing, respectively), and the value of the observation corresponds to their age in days at the time of the event.

Only one indicator is monitored for Empty Sows, the *Number of sows empty longer than 4 days*, per week. Empty or non-productive are defined as any day in the productive cycle on which the sow is not pregnant or lactating. This is a weekly time-series where every week a new observation is recorded, representing the number of sows which, at the beginning of that week, have been empty for 4 days or more. This target value was defined by Swedish stakeholders.

The production phase called Services includes four indicators. The first, *Reservices per week*, is the number of times a week a second artificial insemination was done because the first one was unsuccessful. The second indicator, *Time to reservice*, is the time in days between the service (first artificial insemination) and reservice. Both of these indicators are based on the occurrence of a reservice, but the first one is a weekly time-series (one observation for every week), while *Time to reservice* is a continuous time-series, that is, every reservice event represents a new observation. Failure to conceive after insemination was further monitored by two indicators of weekly observations. The *Percentage of reservices after*

4 weeks was calculated as the percentage of all sows serviced in a given week, which in any of the following 4 weeks went back to heat and were therefore reserviced. Sows which were removed from production within the 4 weeks following a service, and for which the type of exit recorded was not “sold pregnant”, were considered to have been removed due to reproduction failure. The number of reservices plus the number of exits due to failure are considered the total number of failed services, and used as the numerator for the indicator named *Percentage of failures after 4 weeks*.

The category Pregnancy has three indicators. In the first one, called *Pregnancy length*, observations represent every unique farrowing event, with the values representing the number of days from insemination to farrowing. The next indicator, *Time to abortion*, records a new observation for every documented abortion case, with the observation number being the gestation length, in days, from insemination to the observed day of the abortion. The final indicator of this category is *Abortions per week*. It is a weekly time-series where every week a new observation is recorded, representing the number of abortions.

The next production phase, Farrowing, includes five indicators, all of which are continuous time-series (a new observation is recorded for every farrowing event). In *Days between farrowings* the observation value is the number of days since the last farrowing of the same sow. The second indicator, *Total piglets per farrowing*, was created by counting the number of piglets born per farrowing and the third one, *Live piglets per farrowing*, is the number of piglets born alive per farrowing. The next indicator, *Percentage of dead piglets per farrowing*, was constructed by dividing the number of piglets born dead per farrowing by the total number of piglets born in the same litter, multiplied by 100. The last indicator of this production phase, *Mummified piglets per farrowing*, is the number of mummified piglets born per farrowing.

The production phase called Weaning has six indicators. The first one, *Days to weaning*, is the time, in days, from farrowing to weaning of each litter (one observation per weaning event). This indicator depends on the piglets' age at which the farmer decides to wean. The indicator *Litters weaned per week* is also a reflection of the farm's baseline activity and management practices, providing an overview of how many new litters are part of the farm per week. The next two indicators are the number of *Piglets weaned per week* and *Piglets weaned per litter*. Although both the number of pigs farrowed and weaned are registered per sow, there can be transfers of piglets between sows after birth to even out the number of piglets nursed by each sow (cross-fostering). The number of piglets weaned per sow can therefore not be used to calculate, directly, the deaths occurred between farrowing and weaning. Thereby, *Deaths in weaning piglets* was calculated as a total for the farm *per week*, and calculated as the total weaned in the farm in a given week, minus the total born alive in all litters being weaned that week. The last indicator, *Weaning weight per litter (average per piglet)* is the average weight per piglet, in kilograms, of each weaned litter. It was constructed

by dividing the total weight in kilograms of each weaned litter by the number of piglets weaned from that litter.

The next production phase is Post-Weaning. Piglets are often not sold after weaning, and stay at the reproducing farm for a period of growth post-weaning. During this period mortality may be registered in WinPig as number of deaths observed by days, without any information about litters. Therefore, the only indicator that could be calculated was the total *Deaths in weaners per week*.

Lastly, the final category Exit has two indicators. The first indicator, *Last event before exiting per week* groups the exits according to the type of event that preceded the sows' exit: birth, service, reservice, abortion, farrowing or weaning. This indicator is useful for farm management. The last indicator, *Dead sows per week* is the number of sows that died per week.

In some indicators an aberration would be perceived as an increase in the observations, the focus being on detecting those raises. However, in other indicators, an aberration would cause a decrease in the observations and, therefore, aberration detection will have to focus on detecting decreases, rather than increases. This has been indicated in Table 2 in the column "Limit(s)" by a UCL (upper control limit), LCL (lower control limit), or both. A UCL (the minimum value that would have generated an alarm in each data point) is used for detecting increases in the data, while a LCL (maximum number of events that would have caused an alarm in each data point) is applied to follow decreases. When both increases and decreases in the data were intended to be detected, both UCL and LCL were applied.

Table 2: Indicators' categorization and characterization.

Category	Indicators	Time-series	SyS	Limit(s) ^a	Utility
Gilts	Age at first service	Continuous	No	---	Management
	Age at first farrowing	Continuous	No	---	Management
Empty Sows	Number of sows empty longer than 4 days	Weekly	Yes	UCL	Health
Services	Reservices per week	Weekly	Yes	UCL	Health
	Time to reservice	Continuous	Yes	UCL LCL	Health
	Percentage of reservices after 4 weeks	Weekly	Yes	UCL	Health
Pregnancy	Percentage of failures after 4 weeks	Weekly	Yes	UCL	Health
	Pregnancy length	Continuous	Yes	LCL	Health
	Time to abortion	Continuous	No	---	Health
Farrowing	Abortions per week	Weekly	Yes	UCL	Health
	Days between farrowings	Continuous	Yes	UCL LCL	Health
	Total piglets per farrowing	Continuous	Yes	UCL LCL	Health
	Live piglets per farrowing	Continuous	Yes	LCL	Health
	Percentage of dead piglets per farrowing	Continuous	Yes	UCL	Health
Weaning	Mummified piglets per farrowing	Continuous	Yes	UCL	Health
	Days to weaning	Continuous	No	---	Management
	Litters weaned per week	Weekly	No	---	Management
	Piglets weaned per week	Weekly	Yes	LCL	Health
	Piglets weaned per litter	Continuous	Yes	LCL	Health
Post-Weaning	Deaths in weaning piglets per week	Weekly	Yes	UCL	Health
	Weaning weight per litter (average per piglet)	Continuous	Yes	LCL	Health
	Deaths in weaners per week	Weekly	Yes	UCL	Health
	Last event before exiting per week	Weekly	No	---	Management
	Dead sows per week	Weekly	Yes	UCL	Health

SyS – Syndromic Surveillance; Continuous – Continuous time-series; Weekly – Weekly time-series;

UCL – Upper control limit; LCL – Lower control limit.

^a Limit (UCL or LCL) or limits (when both UCL and LCL are used) are only applicable when syndromic surveillance was applied, for monitoring data in comparison to the farms' own historical baselines.

5.3. Time-series analysis

Data organization into indicators resulted in the creation of all the time-series listed in Table 2. As detailed above and indicated in Table 2, besides the discrete time-series commonly used in SyS, many of the indicators were better represented as continuous time-series. The steps of time-series analysis for discrete time-series are well documented in literature, as previously reviewed, and the application of these methods to the discrete time-series is described in this section. Time-series analyses of the continuous time-series was investigated to the extent possible, and is also described in each step.

5.3.1. Retrospective analysis

The presence of temporal effects, and the possible need of pre-processing to remove them was evaluated for all weekly time-series indicators using the R package `{vetsyn}` (Dórea, Widgren, et al. 2015). The package fits regression models to the data using sine and cosine waves to represent seasonal effects, and returns goodness-of-fit measures (Figure 3). These models assume that observations represent specific time units (discrete time-series), and further investigation is needed to understand how a model to remove temporal effects could be applied to the continuous time-series, when the observations are not grouped by a specific temporal unit. Considering the results of the analyses using the `{vetsyn}` package, and the challenges related to the continuous time-series, pre-processing was ultimately discarded from the analysis process, and methods will be presented for all time-series analysed without pre-processing.

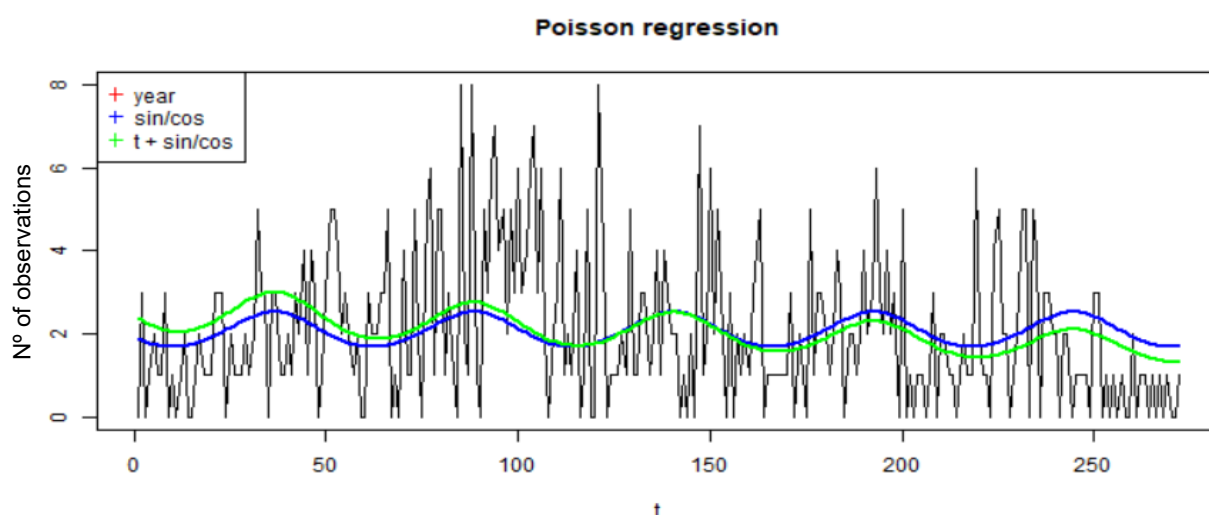


Figure 3: Evaluation of the need to carry out pre-processing in the *Reservices per week* indicator using the R package `{vetsyn}`.

5.3.2. Create a baseline

Exploratory analysis of the time-series showed that event recording was considerably low in the first few years of data. Whether this truly represents a development of the scale of production in the farm, or an adaptation period from moving from paper-based management to digital recording, it was judged that these data were not an accurate training period for the baseline behavior of the various indicators. For weekly indicators the first 200 weeks were removed, with 271 weeks remaining. For the continuous indicators, a maximum number of the most recent 5500 observations per indicator were left.

To create an outbreak-free historical baseline for each indicator in which SyS was applied, a non-parametric method based on moving percentiles was chosen. A function available on R package {vetsyn} (Dórea, Widgren, et al. 2015) was adapted in order to work for both continuous and discrete time-series present in this study. A 95th percentile was selected as the upper limit and a 5th percentile as the lower limit, applying in a moving window of 104 weeks (two years) for weekly time-series. For continuous time-series, we aimed to use the mean number of observations within two production cycles. As the median number of days between farrowings was observed to be 154, we looked at mean number of events, for each indicator, within a 308 (154×2) days window, and used that number as the percentile window. Depending on each indicator's limit(s) (Table 2) any observations falling outside the upper or the lower limits, or both, were removed and substituted by the percentile value itself.

5.3.3. Aberration detection

In the absence of significant temporal effects that needed to be modelled and extracted from the discrete time-series data, control charts were used as the methods of aberration detection. While further research is needed to investigate how temporal effects can be studied in the continuous time-series data, these data more closely resemble the industrial processes which control charts were originally intended to monitor, and therefore control charts were also the natural choice of TADA for these series.

The challenge of applying control charts in VSS context is that each type of chart is most efficient at capturing a specific signal shape (Yahav et al. 2011). As the project aims to detect early signs of disease introduction and management errors, both one-time spikes and slow increases/decreases in the data are important to be detected. As identified in the literature review, the best choice of control charts to detect these two aberration shapes are the Shewhart and EWMA control charts, respectively. Moreover, Dórea et al. 2014 previously demonstrated that EWMA and Shewhart control charts showed complementary performance

in detecting outbreaks of different shapes, without any improvement if CUSUM charts were added.

5.3.4. Algorithms' parameterization and implementation

The detection algorithms chosen, EWMA and Shewhart, were both applied to each indicator. The values used to parameterize the algorithms are listed in Table 3 for both discrete and continuous time-series.

The package {qcc} (Scrucca 2004) provides an original implementation of control charts in R. The package {vetsyn} (Dórea, Widgren, et al. 2015) provides a recursive implementation of the original {qcc} codes. The user chooses the time-points to be evaluated (evaluate window), the number of points to be used as a baseline (baseline window) and the number of time-points to leave between those two (guard-band). The original implementations of Shewhart and EWMA from {qcc} are then applied one time-point at a time within the evaluate-window. The first time-point is evaluated, the result of aberration detection is recorded for that time-point, and the function then moves to the next time-point, with the baseline and guard-band moving accordingly along the time-series.

For the weekly indicators, a baseline of 104 weeks (2 years of data) was chosen based on the visual assessments of real data as suggested in Dórea, McEwen, McNab, Revie, et al. 2013. To avoid baseline contamination, the 2 weeks of guard-band applied as default in the {vetsyn} package for weekly data was maintained. As the analyses done in this project were retrospective, for the indicators that are discrete weekly time-series, the number of observations to apply syndromic surveillance (evaluate window) were the last 165 weeks of the data (Table 3). This number is the difference between the total number of observations present in the weekly data (271) and the baseline (104) and the guard band (2) chosen.

The {vetsyn} workflow was designed specifically for discrete time-series analysed in the SyS context. Their application to the continuous time-series will require further investigation of how to define what constitutes the baseline, guard-band and evaluate windows, and how to move them recursively over continuous (as opposed to discrete) observation points. It was beyond the scope of this work to address this problem, and further research will be needed in the future to adopt the prospective SyS framework to continuous time-series. In the work presented in this document, focus was given to creating the base for this research by applying already available methods to the continuous time-series indicators, and documenting challenges and opportunities. Among the available frameworks, the original implementation of control charts in the {qcc} package is the most suitable for continuous time-series. In that implementation, there is no separation between baseline and evaluate window. The entire

historical series is used to calculate the central, upper and lower control limits as shown in Figure 1. All observations in the time-series are then evaluated against these limits.

In both implementations – {vetsyn} and {qcc} – the user then needs to parameterize the control chart detection function. For EWMA, a smoothing coefficient called lambda is needed. Smaller values of lambda are recommended for detecting smaller shifts, and larger values for larger shifts. Based on common examples from the VSS literature, a value of 0.2 was the one applied.

For the choice of detection limits, the approach suggested in Dórea, McEwen, McNab, Sanchez, et al. 2013 of using a scoring system to combine multiple detection limits was followed. These limits were based on standard deviation thresholds, that is, the number of standard deviations (s.d.) above or below the mean that decide when an observation is abnormal. For both discrete and continuous time-series three different standard deviations (2.5, 3, and 3.5) were provided (Table 3). The 3 standard deviation limit had been already tested with good performance in using production data to detect PRRS outbreak in the United States (Silva et al. 2017). To apply the scoring approach, the standard deviations of 2.5 and 3.5 were added, keeping 3 as the center. The indicators on which syndromic surveillance was applied, were monitored against a UCL, or LCL, or both, according to the information that we wanted to extract from the indicators (Table 2). For each standard deviation limit provided, one UCL was set when we were interested in detecting increases in the observations or/and one LCL when we intended to follow decreases. Thereby, multiple control limits were created. As a result, an alarm score system was created referring to how many standard deviations have been exceeded, as illustrated in Table 3. If for instance an observation was greater than the 3 s.d. limit, but lower than 3.5 s.d., the result was a detection score of 2 (2 detection limits exceeded). For an indicator where we wanted to detect decreases in the observations, with a LCL limit on Table 2, if an observation was greater than 3.5 standard deviations below the mean, an alarm score of -3 was given. Table 3 also shows how alarms appear on the graphs. They are presented in two different colors (red and purple for UCL and LCL alarms, respectively), three sizes according to the scores, and two distinct shapes (a circle for EWMA and an asterisk for Shewhart). The UCL and/or LCL lines shown in the graph correspond to the middle limit threshold (3) relative to the EWMA algorithm.

It is important to keep an outbreak-free baseline after possible outbreaks being detected. The R package {vetsyn} allows the algorithm to correct the data by removing the abnormal observations already detected, and substituting them by the detection limit chosen (UCL/LCL). Dórea, McEwen, McNab, Sanchez, et al. 2013 suggested choosing one algorithm to perform baseline correction, even if multiple are used in parallel for signal detection. Informed by previous experience of senior authors based on the implementation in practice of other systems (Dórea et al. 2014), EWMA was chosen as better able to provide sensitive

correction without over smoothing due to detection of false alarms. Only afterwards, with an outbreak-free baseline, the Shewhart algorithm was applied. As this is a feature of {vetsyn} and therefore not available in {qcc}, we could not implement it on the continuous time-series.

Table 3: Description of the arguments used for each algorithm and R package chosen.

Weekly time-series							Continuous time-series						
<i>Algorithms and R package used</i>	<u>EWMA {vetsyn}</u>			<u>Shewhart {vetsyn}</u>			<u>EWMA {qcc}</u>			<u>Shewhart {qcc}</u>			
<i>Evaluate window^a</i>	165 weeks			165 weeks			All data ⁱ			All data ⁱ			
<i>Baseline window^b</i>	104 weeks			104 weeks			All data ⁱ			All data ⁱ			
<i>Guard-band^c</i>	2 weeks			2 weeks			---			---			
<i>Lambda^d</i>	0.2			---			0.2			---			
<i>Limits s.d.^e</i>	2.5	3	3.5	2.5	3	3.5	2.5	3	3.5	2.5	3	3.5	
<i>Correct baseline^f</i>	True: 3 s.d.			False			False			False			
<i>UCL and LCL lines^g</i>	3 s.d.			---			3 s.d.			---			
<i>Alarms^h</i>	UCL												
	Score	1	2	3	1	2	3	1	2	3	1	2	3
	LCL												
	Score	-1	-2	-3	-1	-2	-3	-1	-2	-3	-1	-2	-3

s.d. – Standard deviation; UCL – Upper control limit; LCL – Low control limit.

^aEvaluate window is number of observations to apply syndromic surveillance.

^bBaseline window is the length of the baseline used to train the algorithm.

^cGuard-band is a gap between the baseline (training data) and the data being evaluated.

^dLambda is the smoothing parameter for EWMA which determines the relative weight of current data to past data.

^eLimits s.d. are the standard deviation limits chosen, that is, the number of standard deviations above or below the mean that decided when an observation was abnormal.

^fCorrect baseline: if True, the algorithm can also be used to correct the data by removing the abnormal observations already detected and substituting them by the limit of the standard deviation of the prediction provided by the algorithm.

^gUCL and LCL lines are the standard deviation limit chosen to appear on the graph.

^hAlarms could be UCL or LCL, or both depending on if the indicator was conceived to detect increases or decreases in the expected number of observations, respectively. The alarms have 3 sizes (scores) according to the Limits s.d. exceeded.

ⁱAll data included in historical dataset, which for the continuous time-series indicator is at most the last 5500 observations.

5.3.5. Evaluation of performance in a practical scenario

The next step was to test the implemented syndromic surveillance system with an outbreak of an infectious disease. No information concerning the animal health of the farm under study was given, but no known outbreaks of regulated diseases were known to be present in the historical data. In this case, it is common practice in SyS to create simulated data by designing artificial outbreaks and then inject them into the data (Dórea, McEwen, McNab, Revie, et al. 2013).

It was of specific interest to inject outbreaks of Porcine reproductive and respiratory syndrome (PRRS) since Sweden is free of the disease, despite being endemic in Europe (Carlsson et al. 2009). Also, and as reported by Lunney et al. 2010, it is among the diseases with the highest economic impact in modern pig production worldwide (consult appendix for more information about PRRS). To assess whether the SyS system would be able to detect PRRS emergence in the Swedish farm under study, the first step was to choose which of the health-related indicators created would experience changes during a PRRS outbreak. Based on Pejsak and Markowska-Daniel 1997; Valdes-Donoso et al. 2018, the indicators listed in Table 4 were deemed most suitable to inject outbreak signals. Outbreak signals were assumed to follow a log-normal curve, when possible, which is the typical distribution for the rise and fall in the number of infected animals in a typical infectious disease model (Keeling and Rohani 2008). To simulate a log-normal, we reviewed the animal health and reproduction effects reported during the PRRS outbreaks that occurred in the literature mentioned, as described in Table 4. For each indicator, the artificial outbreaks were injected on the historical data available as explained in Table 4. Outbreaks were assumed to always start in week 1 and last 13 weeks, being therefore confined to a single quarter of the year. The time-series for each indicator, discrete or continuous, from the second quarter of 2015 until the first quarter of 2018 (3 years, 12 quarters in total) were used as the basal farm behavior. Twelve copies of each indicator series were created, and a single outbreak signal was injected in each of those copies, always in a different quarter. In total, 120 outbreaks were injected in the 10 indicators during 3 years.

Table 4: Information used to inject artificial outbreak signals of PRRS, and explanation of how it was done, in each one of the 10 indicators chosen.

<i>Indicators</i>	<i>Values reported during PRRS outbreak^a</i>	<i>Outbreaks' injection explanation</i>	<i>Ref.</i>
Reservices per week	Increased at week t3 with maximum value at t8 (tripled), then decreased until t28.	The log-normal curve was constructed based on the increase in magnitude reported: starting at a value of 0, the curve was modelled to start raising on week 3, until reaching a magnitude of 2 on week 8, and then decrease slowly until reaching 0 again on week 28, but only values up to week 13 were considered. The magnitude values from the curve were then multiplied by the baseline mean in the corresponding weeks. Next, the values obtained were added to the indicators' original data.	A
Pregnancy length	From t1 to t4: 7% of farrowed sows had premature farrowing ^b ; from t5 to t8: 26% of farrowed sows had premature farrowing; from t9 to t12: 8% of farrowed sows had premature farrowing.	The percentage of premature farrowings in each week reported in the study were used to draw a random sample of all farrowings recorded on the farm in the corresponding weeks. Drawn farrowings were transformed into premature farrowings by replacing their recorded pregnancy length in days with a value randomly drawn from an uniform distribution between 105 and 110.	B
Abortions per week	Doubled at week t1 and quintupled at week t2, then decreased during 20 weeks.	Based on the increase in magnitude reported, a log-normal curve was created to represent the probability of an abortion in each week under an outbreak scenario. Abortions were a very rare event in the original data, therefore, these probabilities were used to choose in which weeks to inject one abortion caused by the outbreak, and the results were added to the original data.	A
Live piglets per farrowing	Decreased 8% between t3 and t20, biggest decline at t4 and t5.	The log-normal curve was constructed based on the magnitude decrease reported, and then it was multiplied by the original data. Next, the values obtained were subtracted to the indicators' original data.	A
Percentage of dead piglets per farrowing	Assumed to have an opposite behaviour to the indicator <i>Live piglets per farrowing</i>	A symmetric curve to the indicator <i>Live piglets per farrowing</i> was created, and then it was multiplied by 100. Next, the values obtained were added to the indicators' original data.	
Mummified piglets per farrowing	From t1 to t4: 3% of the total piglets farrowed were mummified; from t5 to t8: 22% of the total piglets farrowed were mummified; then decreased until t44.	The log-normal curve was constructed based on the values reported, and then it was multiplied by the total number of piglets farrowed. Next, the values obtained were added to the indicators' original data.	B

Indicators	Values reported during PRRS outbreak^a	Outbreaks' injection explanation	Ref.
Piglets weaned per week	Decreased from week t1 to week t7 (less 23%), then increased until t35.	The log-normal curve was constructed based on the magnitude decrease reported, and then it was multiplied by the original data. Next, the values obtained were subtracted to the indicators' original data.	A
Piglets weaned per litter	Assumed to have the same behaviour as the indicator <i>Piglets weaned per week</i>	The log-normal curve was constructed based on the magnitude decrease reported, and then it was multiplied by the original data. Next, the values obtained were subtracted to the indicators' original data.	
Deaths in weaning piglets per week	At t2 increased 26% and at t3 47%, then decreased until t12.	The log-normal curve was constructed based on the magnitude increase reported, and then it was multiplied by the original data. Next, the values obtained were added to the indicators' original data.	A
Dead sows per week	Increased 10% in week t3, then decreased until t7.	The log-normal curve was constructed based on the magnitude increase reported, and then it was multiplied by baseline mean. Next, the values obtained were added to the indicators' original data.	A

Ref – Reference; A – (Valdes-Donoso et al. 2018); B – (Pejsak and Markowska-Daniel 1997);

^a Outbreak began at week t1.

^b sows that give birth before the 110th day of pregnancy.

Note that in Table 4 in the column “Outbreaks' injection explanation” in some indicators the log-normal curves were multiplied by the baseline mean and in others by the original data. Multiplication by the original data was performed, when possible, to preserve original noise and variability. However, when zero was a common value on the indicators' original data, the log-normal curves were multiplied by the baseline mean to avoid losing the log-normal curves expression, which would have happened if they were multiplied by zero.

Performance of the control charts was evaluated based on three main performance measures:

- False alarm rate – the false alarm rate was evaluated based on the number of alarms triggered when the control charts are applied to the original observed data, without any injected outbreaks. This assumes that the original data does not contain any major disease outbreaks – although it may contain unspecific sparse health problems, diffuse management errors, and other sources of random noise. These sources of noises are expected to create a basal rate of alarms that do not deserve particular attention (false alarms). In the VSS literature it is suggested that algorithms should be parameterized for a maximum false alarm rate of 3% (Mandl, Reis, et al. 2004).
- Sensitivity – evaluated based on the control charts ability to detect the injected outbreaks.

- Timeliness – evaluated based on how early the injected outbreaks are detected, depending on the type of control charts for each of the indicators evaluated.

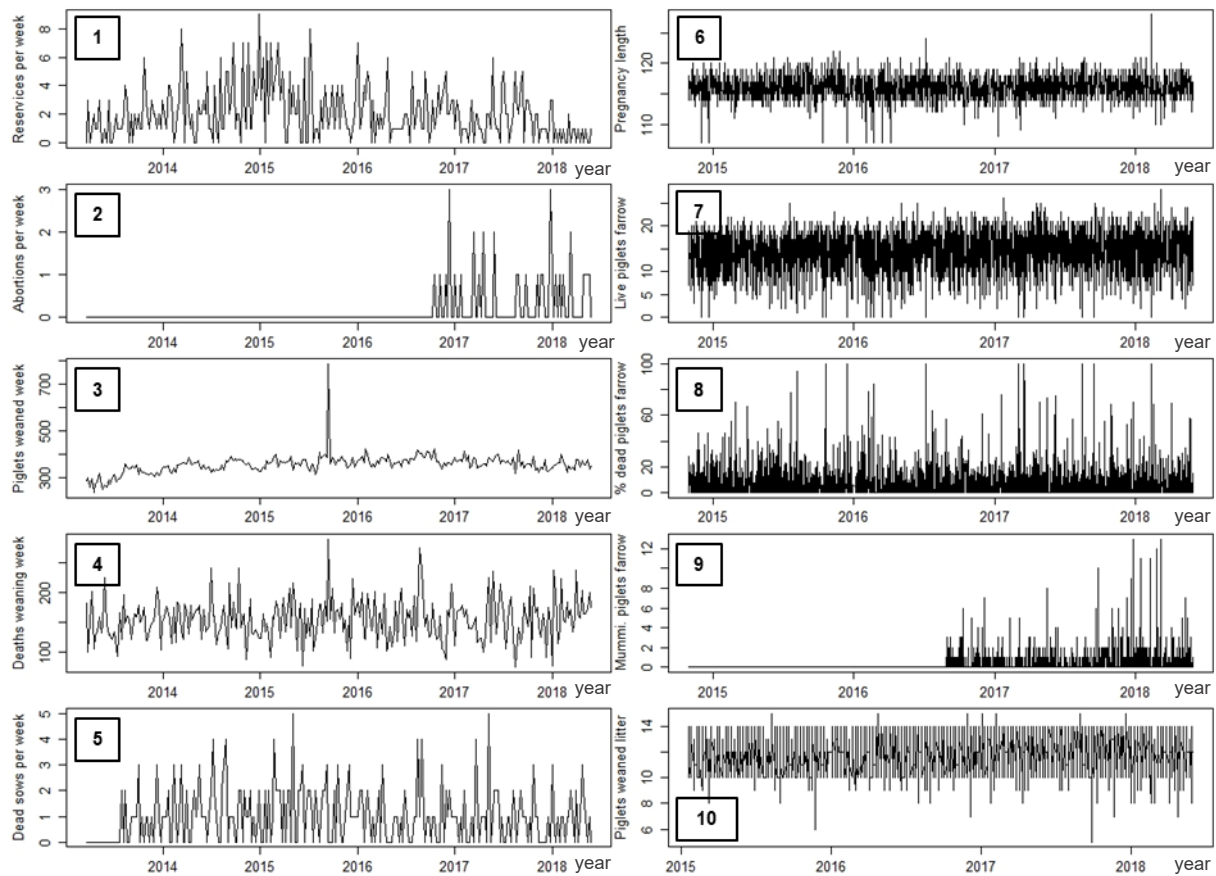
5.4. Present results in interface

It was important to present a user-friendly dashboard where the stakeholders could easily access. In that interface, they could look at the results of the implemented SyS system on the farm, and also check the other indicators that they were interested in keeping for production and reproduction management. Therefore, an early dashboard version was constructed using R package {flexdashboard} (Iannone et al. 2020).

6. Results

6.1. Time-series analysis with focus on temporal aberration detection

As addressed before, the SyS system created was assessed using simulated data with injected PRRS outbreaks. Therefore, the presentation of results will be focused on those indicators chosen for injection of the simulated outbreaks (Table 4). Figure 4 illustrates the original observed values of those ten indicators. It is noteworthy that the indicator *Abortions per week* ranges only from 0 to 3, and up to more than half of the time-series the values were 0. Moreover, in the indicators *Pregnancy length* and *Piglets weaned per litter* the values mostly range between 114 and 120 days, and between the 10 and 14 days, respectively.



1 – Reservices per week; 2 – Abortions per week; 3 – Piglets weaned per week; 4 – Deaths in weaning piglets per week; 5 – Dead sows per week; 6 – Pregnancy length; 7 – Live piglets per farrowing; 8 – Percentage of dead piglets per farrowing; 9 – Mummified piglets per farrowing; 10 – Piglets weaned per litter.

Figure 4: Time-series of weekly (left) and continuous (right) observed values from the ten indicators before the PRRS outbreaks injection.

6.1.1. Weekly time-series

Table 5 shows the total number of weeks on which SyS was applied, and of those weeks how many triggered an alarm, for each alarm score, on the original data (before the outbreaks injection). The false alarm rate was calculated by dividing the total number of weeks with alarms (any alarms, therefore based on the lowest detection threshold) by the total number of weeks evaluated. Every alarm, to be an alarm, had exceeded the first limit score. Of these, some exceeded score 2/-2, and of these, in turn, also exceeded score 3/-3.

Regarding the EWMA algorithm, the indicators with the lowest false alarm rate were *Reservices per week* and *Dead sows per week*, both with 0.6%. In opposition, the indicator with the highest false alarm rate was *Abortions per week* with 13.3%.

Concerning the Shewhart control chart, the indicators *Reservices per week* and *Piglets weaned per week*, both had a 0% false alarm rate. The indicator with the highest false alarm rate was again *Abortions per week*, also with 13.3%.

Table 5: Number of alarms triggered on the original data per weekly indicator for both control charts used. Alarms were counted by score, and the false alarm rate was calculated based on the lowest detection threshold (lowest score).

Weekly Indicators	N° of weeks eval.	Control Chart	N° of weeks with alarms			FA rate (%)
			score 1 / -1	score 2 / -2	score 3 / -3	
Reservices per week	165	EWMA	1	1	0	0.6
		Shewhart	0	0	0	0
Abortions per week	165	EWMA	22	22	22	13.3
		Shewhart	22	22	22	13.3
Piglets weaned per week	165	EWMA	9	2	0	5.5
		Shewhart	0	0	0	0
Deaths in weaning piglets per week	165	EWMA	3	1	0	1.8
		Shewhart	7	4	2	4.2
Dead sows per week	165	EWMA	1	0	0	0.6
		Shewhart	2	2	1	1.2

eval. – evaluated; FA – False alarm.

* The scores were counted as positives or negatives according to whether the indicator is monitored based on UCL or LCL, respectively (Table 2).

Figure 5 depicts all the alarms produced by the EWMA control chart on the original data throughout the twelve quarters evaluated.

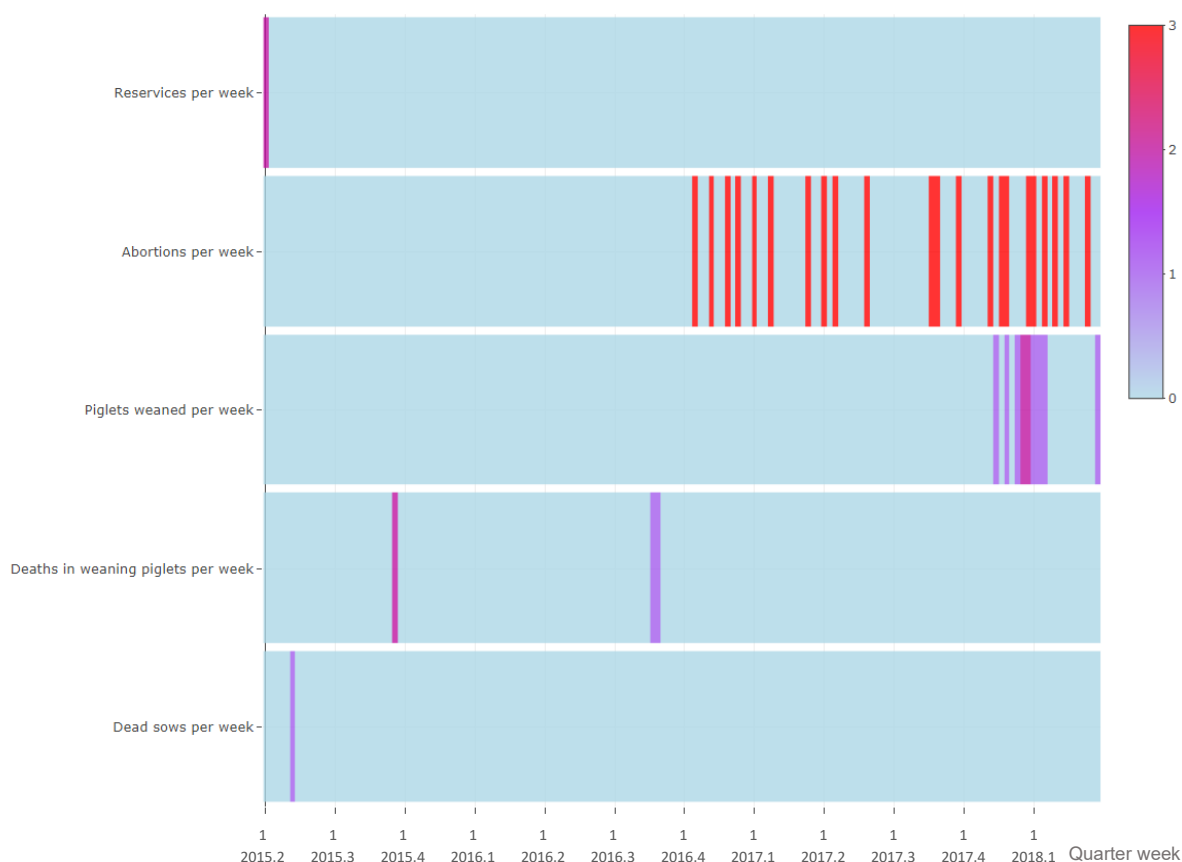


Figure 5: Alarms triggered by EWMA algorithm on the weekly indicators on the original data in the twelve quarters. The color bar ranges from 0 to 3 since a scoring system was applied. If the color bar number is 0, it means that no abnormality was detected. When an alarm was triggered a detection score with a value between 1 and 3 was created, representing the “severity” of the alarm.

*1 2015.2 – Corresponds to the first week of the second quarter of the year 2015. The other X-axis’ values were built using the same logic.

*The indicator *Piglets weaned per week* is a LCL indicator. For these indicators the scores assigned ranges from -1 to -3 (Table 3). In order to be consistent with the other indicators chosen and to facilitate the analysis of the heat map, the scores were labeled from 1 to 3 despite the alarms taken for this indicator were the ones with negative scores.

In Figure 6 it is visible, before the outbreaks’ injection, when the Shewhart alarms were triggered throughout the twelve quarters.

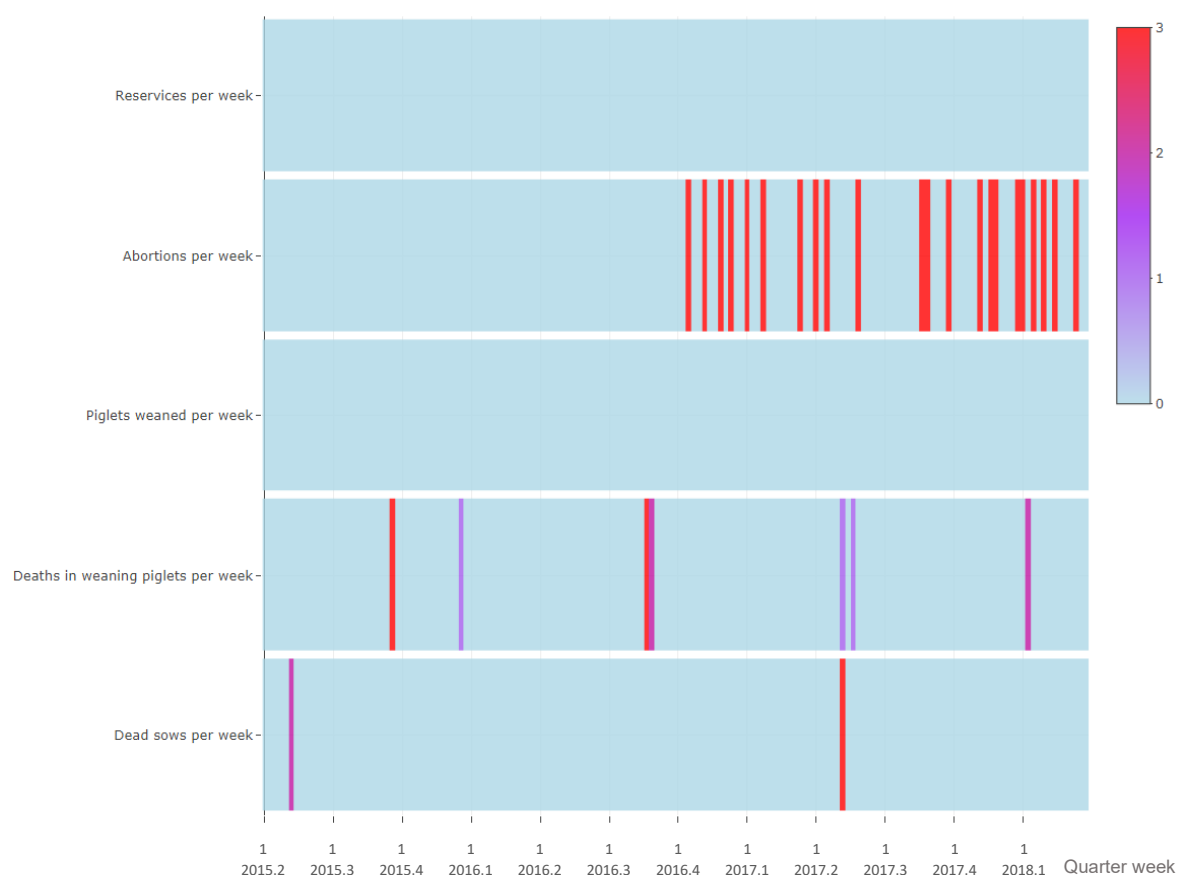


Figure 6: Alarms triggered by Shewhart algorithm on the weekly indicators on the original data in the twelve quarters. The color bar ranges from 0 to 3 since a scoring system was applied. If the color bar number is 0, it means that no abnormality was detected. When an alarm was triggered a detection score with a value between 1 and 3 was created, representing the “severity” of the alarm.

*1 2015.2 – Corresponds to the first week of the second quarter of the year 2015. The other X-axis’ values were built using the same logic.

*The indicator *Piglets weaned per week* is a LCL indicator. For these indicators the scores assigned ranges from -1 to -3 (Table 3). In order to be consistent with the other indicators chosen and to facilitate the analysis of the heat map, the scores were labeled from 1 to 3 despite the alarms taken for this indicator were the ones with negative scores.

Next, the performance of the algorithms in the presence of the injected outbreaks was assessed. Table 6 shows the total number of alarms generated by each of the two control charts for each of the 12 simulated outbreaks (represented by the quarters in which these outbreaks were injected).

Regarding EWMA algorithm and as illustrated in Table 6, in the indicators *Abortions per week* and *Piglets weaned per week* all twelve quarters on which the outbreaks were injected, triggered at least one EWMA alarm. In *Reservices per week* and *Deaths in weaning piglets per week* all quarters also triggered at least one alarm, except for the first quarter of 2018 and the second quarter of 2016, respectively. Lastly, in the indicator *Dead sows per*

week, 7 out of 12 quarters did not generate any alarm. About the total number of EWMA alarms triggered over the 12 quarters, the indicator *Piglets weaned per week* was the one with more alarms generated (78), followed by *Abortions per week* (69) and *Reservices per week* (54). The indicator with less EWMA alarms triggered throughout the 12 quarters was *Dead sows per week* (7).

Concerning Shewhart control chart, the indicator *Abortions per week* was the only indicator where all 12 quarters on which the outbreaks were injected, triggered at least one alarm. For the indicator *Deaths in weaning piglets per week* 11 out of 12 simulated outbreaks triggered at least one alarm, the exception being the outbreak injected on the fourth quarter of 2016. Lastly, the less sensitive indicators were *Reservices per week*, *Piglets weaned per week* and *Dead sows per week*, where 6, 7, and 8 out of 12 simulated outbreaks did not generate any alarm, respectively. About the total number of Shewhart alarms triggered, the indicator *Abortions per week* was by far the one with more alarms generated (69), followed by *Deaths in weaning piglets per week* (32). The indicator with less alarms triggered over the 12 quarters was again *Dead sows per week* (4).

Table 6: Number of alarms triggered by EWMA and Shewhart algorithms per quarter and per weekly indicator, after the outbreaks' injection.

Weekly Indicators	Control Chart	2015.2	2015.3	2015.4	2016.1	2016.2	2016.3	2016.4	2017.1	2017.2	2017.3	2017.4	2018.1
Reservices per week	EWMA	9	4	1	8	1	3	7	2	6	8	5	0
	Shewhart	2	0	0	2	0	1	3	0	2	6	0	0
Abortions per week	EWMA	5	5	7	2	3	6	6	7	6	6	7	9
	Shewhart	5	5	7	2	3	6	6	7	6	6	7	9
Piglets weaned per week	EWMA	4	6	2	5	5	4	5	7	9	7	11	13
	Shewhart	1	3	0	0	0	0	0	0	0	1	9	9
Deaths in weaning piglets per week	EWMA	5	2	4	3	0	5	1	4	2	2	2	1
	Shewhart	4	3	3	2	1	4	0	3	3	2	4	3
Dead sows per week	EWMA	2	0	0	0	0	2	0	0	1	1	1	0
	Shewhart	1	0	0	0	0	0	0	0	1	0	1	1

*2015.2 – Corresponds to the second quarter of the year 2015. The following columns' names were created using the same logic.

Figures 7 and 8 show, for the weekly indicators, the total number of EWMA and Shewhart alarms triggered, respectively, in each simulated outbreak week, for all twelve simulated quarters simultaneously (superimposed). For instance, week 1 shows the sum of alarms triggered in week 1 for each of the twelve simulated outbreak quarters. The log-normal curves used to simulate the outbreaks are also presented on the heat map, allowing evaluation of the timeliness of detection in light of the injected aberrations.

Regarding EWMA control chart and as illustrated in Figure 7, *Abortions per week* was the earliest indicator. Alarms were triggered already in the first outbreak week in 3 (out of 12) of the simulated outbreak quarters, in the second week in 4/12 simulated outbreaks, and in the third week in 7/12. The weeks with the highest number of alarms were weeks six and nine, when EWMA alarms for this indicator were recorded in 8 out of 12 simulated outbreaks. Then, indicators *Reservices per week* and *Piglets weaned per week* had both one simulated outbreak generating 1 alarm in week one. However, in the second indicator, the number of quarters with alarms increases in the following weeks as opposed to *Reservices per week* where the next alarm is only detected in week six. It is noteworthy that in *Piglets weaned per week* all 12 simulated outbreaks generated an alarm in week eleven. Next, the indicator *Deaths in weaning piglets per week* has its first alarm in week three with 4 simulated outbreaks generating alarms in that week, and its highest number of simulations triggering an alarm were 8, in week six. The last indicator *Dead sows per week* was the later indicator triggering alarms, with only 2 simulations generating an alarm in week four.

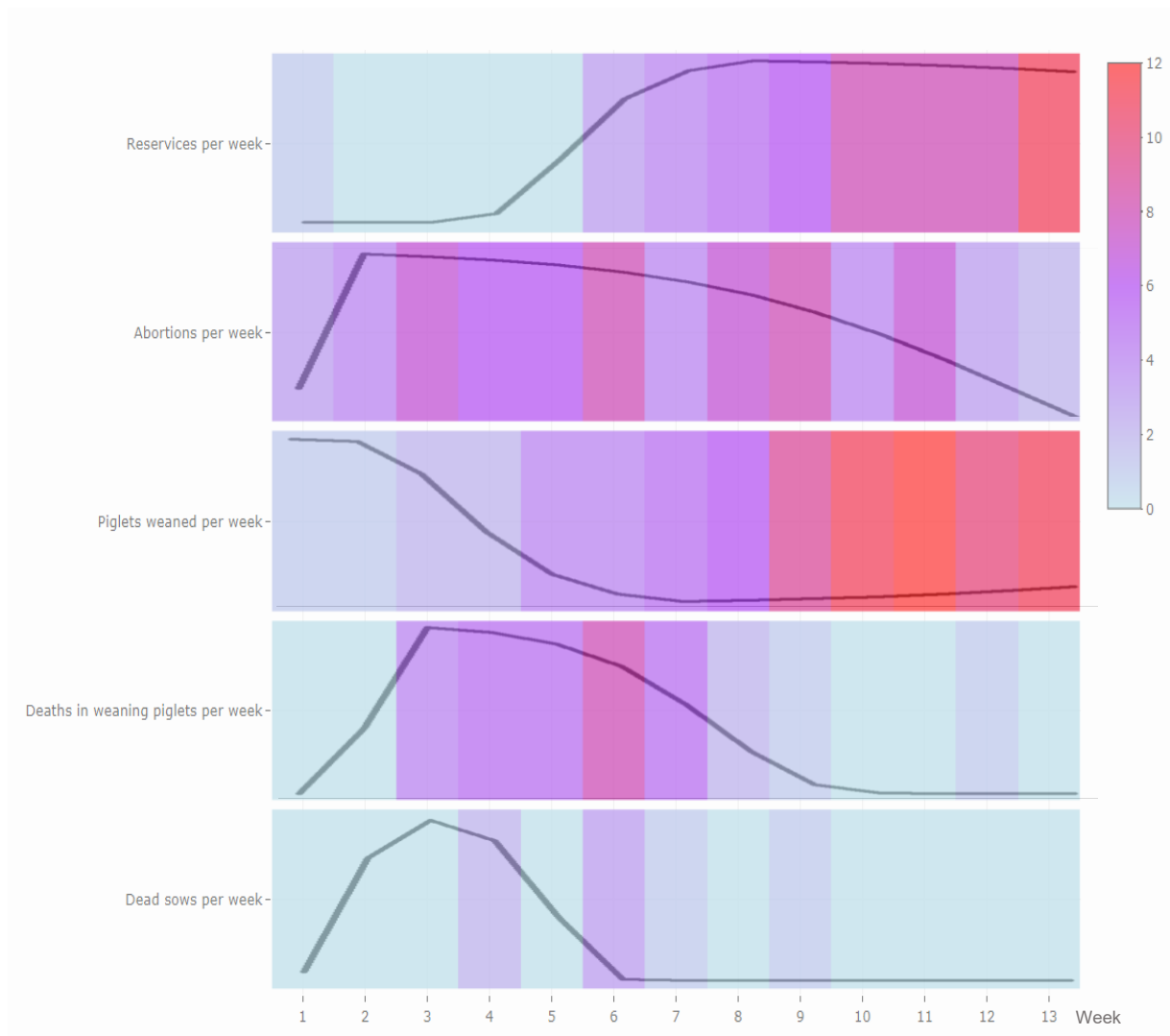


Figure 7: Total number of alarms triggered by the EWMA control chart in each week of the injected outbreaks, superimposed for all twelve quarters simulated, on the weekly indicators. The corresponding log-normal curves used for aberration injection are also depicted.

Concerning the Shewhart algorithm, Figure 8 was created. Note that unlike what happened with Figure 7 for EWMA control chart, in Figure 8 the maximum number of simulated outbreak quarters triggering an alarm was 8. Again, the indicator *Abortions per week* was the earliest, in the first outbreak week 3 out of 12 of the simulated outbreak quarters triggered an alarm, in the second week were 4/12 simulated outbreaks, and in the third week were 7/12. The highest number of simulated outbreaks triggering a Shewhart alarm were recorded as 8 out of 12, in weeks six and nine. The indicator *Dead sows per week* also generated an alarm on week one, with just 1 simulated outbreak generating it. However, this indicator only had more two weeks with alarms, weeks four and six, with 1 and 2 simulated outbreaks triggering them, respectively. Then, the indicator *Deaths in weaning piglets per week* was the third earliest to detect an alarm, with 3/12 simulated outbreaks generating an alarm in week two, 7/12 in week three and 5/12 quarters in week four. The remaining indicators, *Piglets weaned*

per week and *Reservices per week*, generated the first alarm on weeks five (3 out of 12 simulated outbreaks detected) and six (1 out of 12 simulated outbreaks detected) respectively, being *Reservices per week* the later indicator triggering alarms.

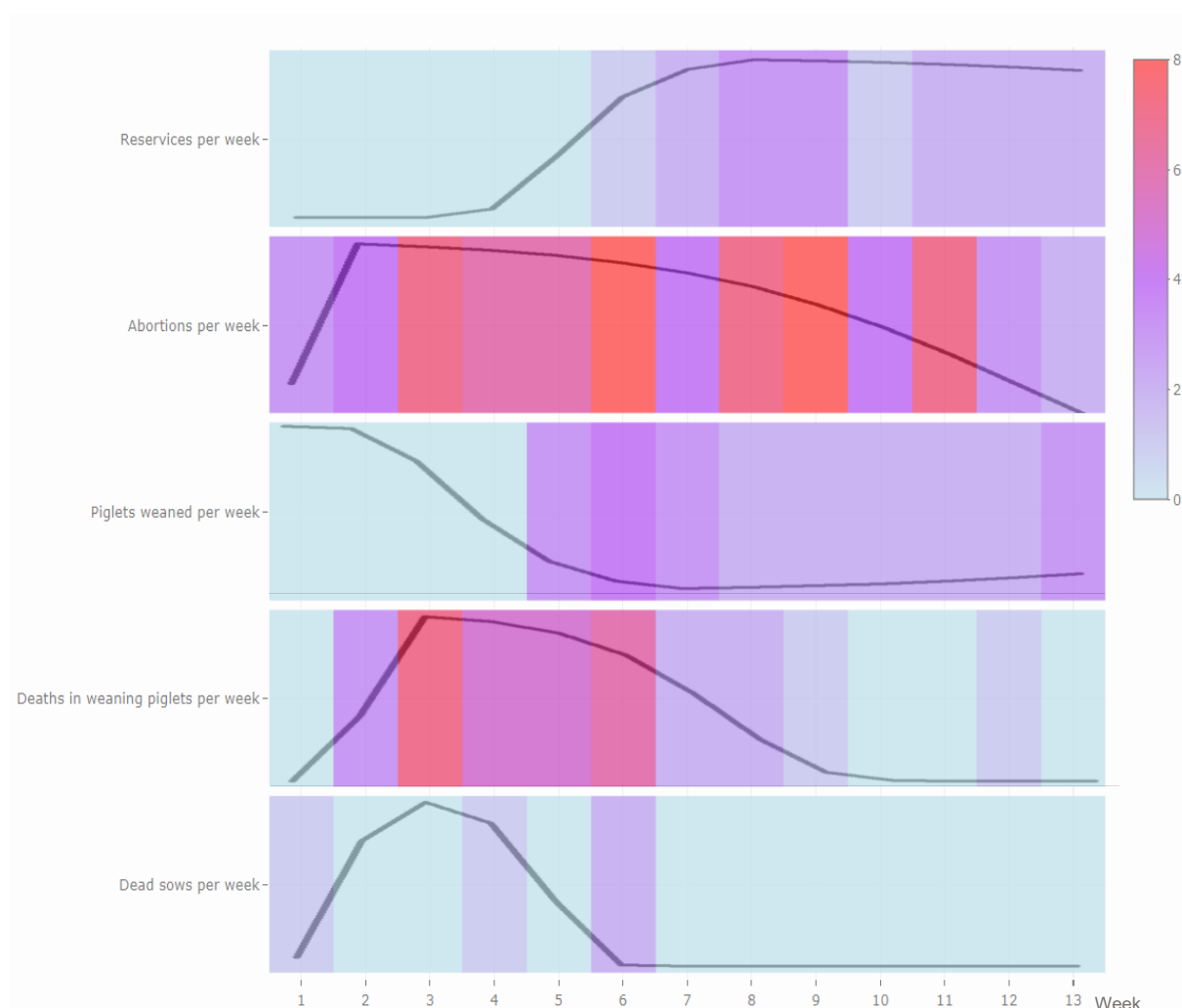


Figure 8: Total number of alarms triggered by the Shewhart control chart in each week of the injected outbreaks, superimposed for all twelve quarters simulated, on the weekly indicators. The corresponding log-normal curves used for aberration injection are also depicted.

6.1.2. Continuous time-series

Table 7 presents, for each of the continuous time-series indicators evaluated, the total number of observations in the original data (without outbreak injection), and of those observations how many triggered an alarm for each of the control charts evaluated, for the three detection limits employed. The false alarm rate was calculated by dividing the total number of observations with alarms (any alarms, therefore based on the lowest detection threshold) for the total number of observations evaluated.

Concerning EWMA control chart performance, the indicator with the highest false alarm rate was *Pregnancy length*, 14.8%. *Mummified piglets per farrowing* and *Piglets weaned per litter* had a false alarm rate of 7.8% and 7.6%, respectively. The indicators with the lowest false alarm rate were *Live piglets per farrowing* and *Percentage of dead piglets per farrowing*, with 2.2% and 3.2%, respectively.

Regarding the Shewhart algorithm, the indicator with the lowest false alarm rate was *Piglets weaned per litter* (0.4%), followed by the indicator *Pregnancy length* with (1%). On the other hand, *Mummified piglets per farrowing* was the indicator with the highest false alarm rate, with 3.1%.

Table 7: Number of alarms triggered on the original data per continuous indicator for both control charts used. Alarms were counted by score, and the false alarm rate was calculated based on the lowest detection threshold (lowest score).

<i>Continuous Indicators</i>	<i>N° of obs. eval.</i>	<i>Control Chart</i>	<i>N° of observations with alarms</i>			<i>FA rate (%)</i>
			<i>score 1 / -1</i>	<i>score 2 / -2</i>	<i>score 3 / -3</i>	
Pregnancy length	4603	EWMA	681	452	275	14.8
		Shewhart	45	21	13	1
Live piglets per farrowing	4603	EWMA	102	42	17	2.2
		Shewhart	96	36	15	2.1
Percentage of dead piglets per farrowing	4603	EWMA	147	102	70	3.2
		Shewhart	104	76	58	2.3
Mummified piglets per farrowing	4603	EWMA	358	336	292	7.8
		Shewhart	142	55	55	3.1
Piglets weaned per litter	4901	EWMA	372	199	93	7.6
		Shewhart	18	4	4	0.4

Obs. – Observations; eval. – evaluated; FA – False alarm.

* The scores were counted as positives or negatives according to whether the indicator is monitored based on UCL or LCL, respectively (Table 2).

Figure 9 shows the alarms detected by the EWMA algorithm for each of the continuous indicators. For these continuous indicators, the X-axis does not represent a specific unit of time, and all the observed events are simply plotted in their order of occurrence.



Figure 9: Alarms triggered by EWMA algorithm on the continuous indicators on the original data in all observations available during the twelve quarters. The color bar ranges from 0 to 3 since a scoring system was applied. If the color bar number is 0, it means that no abnormality was detected. When an alarm was triggered a detection score with a value between 1 and 3 was created, representing the “severity” of the alarm.

*The indicators *Piglets weaned per litter*, *Live piglets per farrowing* and *Pregnancy length* are LCL indicators. For these indicators the scores assigned ranges from -1 to -3 (Table 3). In order to be consistent with the other indicators chosen and to facilitate the analysis of the heat map, the scores were labeled from 1 to 3 despite the alarms taken for these indicators were the ones with negative scores.

In Figure 10 can be seen the Shewhart alarms generated against the total observed events for each indicator, in their order of occurrence.

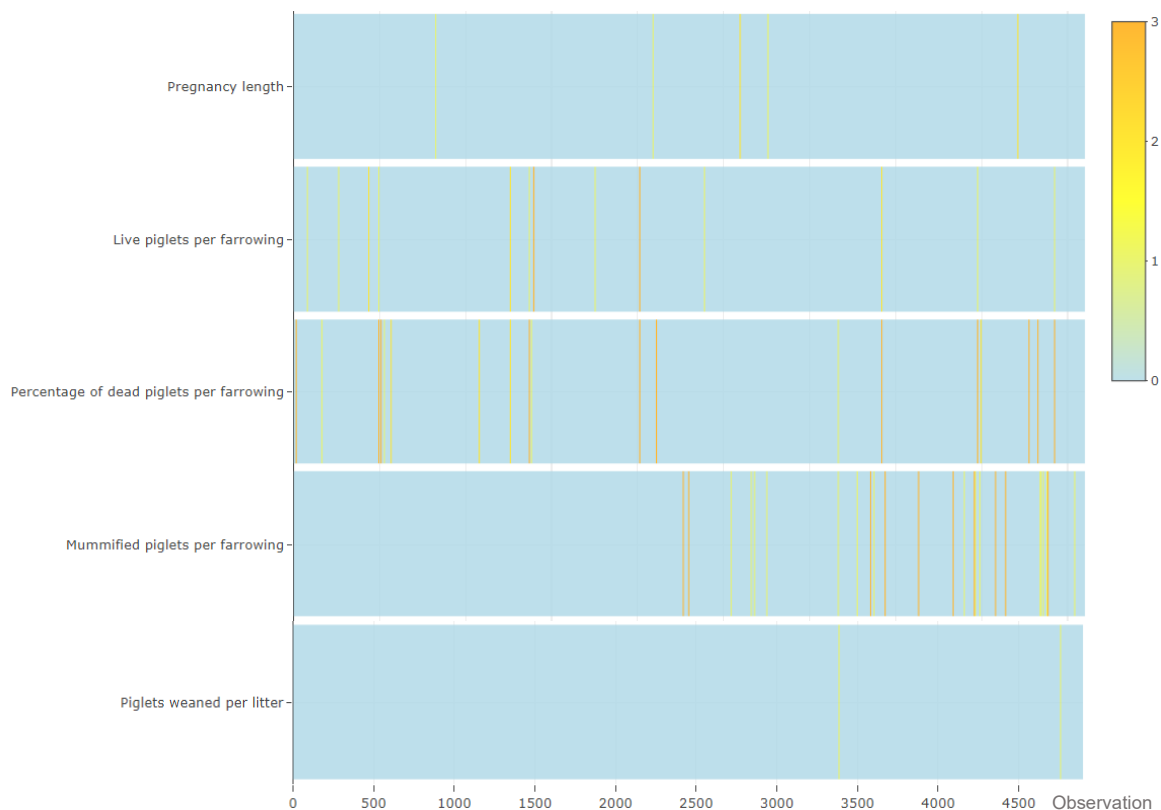


Figure 10: Alarms triggered by Shewhart algorithm on the continuous indicators on the original data in all observations available during the twelve quarters. The color bar ranges from 0 to 3 since a scoring system was applied. If the color bar number is 0, it means that no abnormality was detected. When an alarm was triggered a detection score with a value between 1 and 3 was created, representing the “severity” of the alarm.

*The indicators *Piglets weaned per litter*, *Live piglets per farrowing* and *Pregnancy length* are LCL indicators. For these indicators the scores assigned ranges from -1 to -3 (Table 3). In order to be consistent with the other indicators chosen and to facilitate the analysis of the heat map, the scores were labeled from 1 to 3 despite the alarms taken for these indicators were the ones with negative scores.

Next, the performance of the control charts in the presence of the injected outbreaks was assessed. Table 8 shows the total number of alarms detected for each of the continuous indicators, and for each control chart applied, individually for each of the 12 simulated outbreaks. In all twelve quarters on which the outbreaks were injected, at least two alarms were generated for every indicator, for both control charts.

Regarding EWMA, the indicators *Mummified piglets per farrowing* and *Piglets weaned per litter* were the ones that generated the most alarms in all twelve quarters, with a total of 4565 and 3700 alarms triggered over the 12 simulated outbreaks, respectively. In opposition, *Live piglets per farrowing* was the indicator with less alarms triggered, with only two alarms on

the fourth quarter of 2016 and on the third quarter of 2017, and a total of 307 alarms generated in all 12 quarters.

Concerning Shewhart algorithm and as exhibited in Table 8, *Mummified piglets per farrowing* was again the indicator that triggered the most alarms in all twelve quarters, with a total of 2479 alarms generated. In opposition, the indicators *Live piglets per farrowing* and *Percentage of dead piglets per farrowing* were the indicator with fewer alarms triggered, with 121 and 177 alarms generated throughout the 12 quarters, respectively.

Table 8: Number of alarms triggered by EWMA and Shewhart algorithms per quarter and per continuous indicator, after the outbreaks' injection.

Continuous Indicators	Control Chart	2015.2	2015.3	2015.4	2016.1	2016.2	2016.3	2016.4	2017.1	2017.2	2017.3	2017.4	2018.1
Pregnancy length	EWMA	156	126	115	190	186	206	189	177	141	114	150	129
	Shewhart	47	52	46	52	46	47	47	44	48	43	44	43
Live piglets per farrowing	EWMA	73	32	45	42	45	23	2	17	3	2	14	9
	Shewhart	11	12	8	19	15	12	8	9	6	5	9	7
Percentage of dead piglets per farrowing	EWMA	188	186	152	167	143	169	149	176	168	138	162	190
	Shewhart	28	16	14	19	6	21	10	11	14	12	12	14
Mummified piglets per farrowing	EWMA	389	435	379	390	381	405	365	355	385	365	350	366
	Shewhart	203	229	203	194	197	230	211	205	232	225	210	140
Piglets weaned per litter	EWMA	365	375	373	331	296	332	280	269	271	253	246	309
	Shewhart	77	73	66	68	61	47	50	29	30	41	43	51

*2015.2 – Corresponds to the second quarter of the year 2015. The following columns' names were created using the same logic.

Figures 11 and 12 show, for the continuous indicators, the total number of alarms triggered in each week of the injected outbreaks, for all twelve simulated outbreaks simultaneously (superimposed). The log-normal curves used to simulate the outbreak signal are also presented on the heat map.

To assess the EWMA timeliness and as can be seen in Figure 11, the indicator which triggered the most alarms in an earlier phase was *Mummified piglets per farrowing*, with 319 out of 345 observations generating an alarm in week one, 355/356 observations in week two, and 374/374 observations in week three (all observations triggered alarms). During the thirteen weeks of the simulated outbreak, all weeks had more than 300 alarms triggered in this indicator. The second earliest indicator with more alarms was *Piglets weaned per litter*, with 153/380 observations generating an alarm in week one, 167/373 observations in week two, and 124/379 observations in week three. Then, the number of alarms increased, with the maximum number of 377 out of 377 observations generating an alarm in week eight. *Percentage of dead piglets per farrowing* and *Pregnancy length* had a similar number of alarms triggered, despite the first indicator having more alarms generated in an early phase. However, between weeks five and eight, *Pregnancy length* exceeded the number of alarms. The *Live piglets per farrowing* was the indicator which triggered less alarms in an earlier phase, with only with 5/345 observations generating an alarm in week one, 4 observations in weeks two and three (out of 356 and 374, respectively), and 52/341 observations in week four (maximum number of alarms triggered).

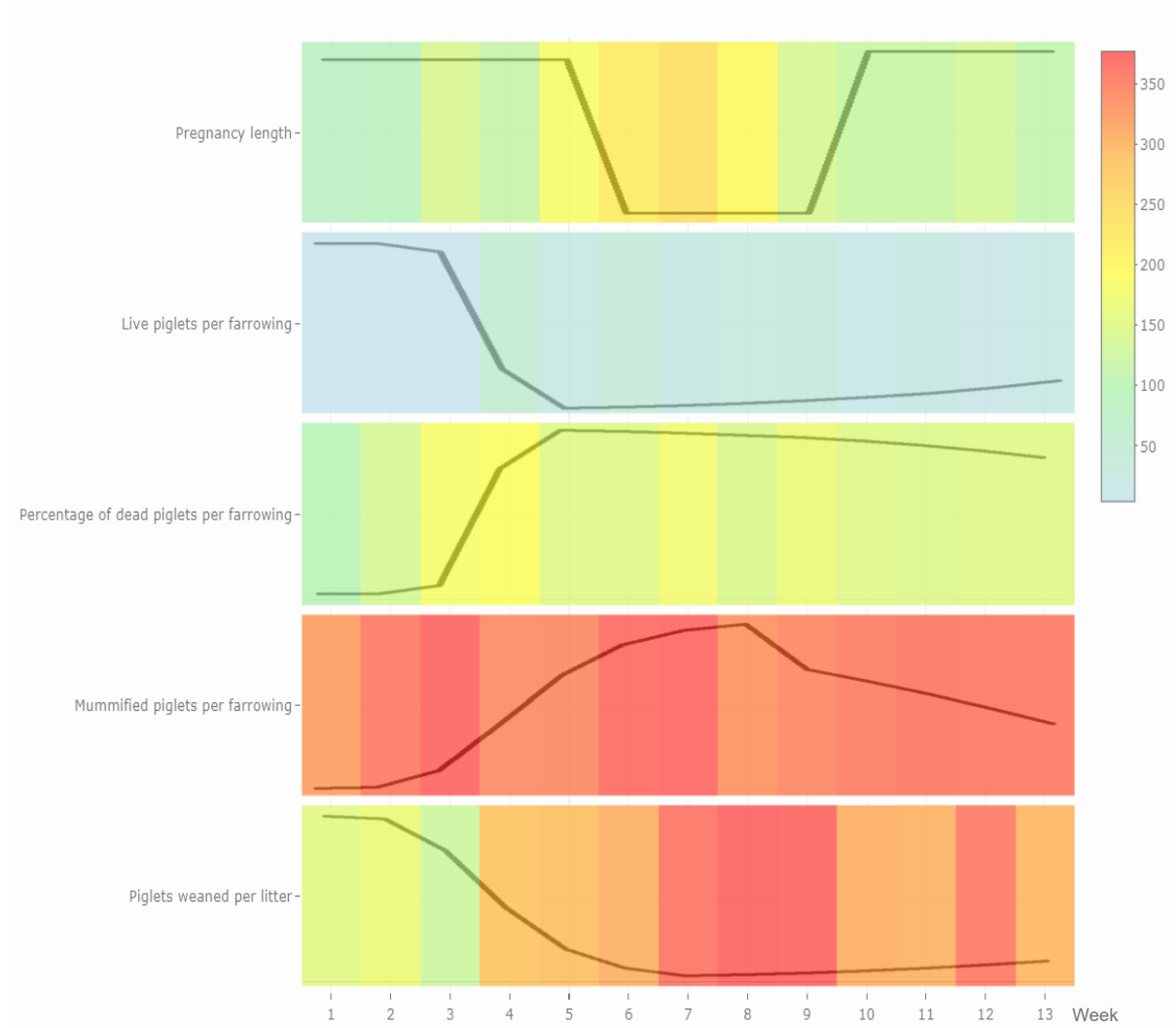


Figure 11: Total number of alarms triggered by the EWMA control chart in each week of the injected outbreaks, superimposed for all twelve quarters simulated, on the continuous indicators. The corresponding log-normal curves used for aberration injection are also depicted.

Concerning the Shewhart timeliness for the continuous indicators, Figure 12 was created. The scale in Figure 12 was mainly influenced by the indicator *Mummified piglets per farrowing*, making it hard to see the results for the other indicators. Analyses of the results showed that the indicator which triggered the most alarms in an earlier phase was *Pregnancy length*, with 19 out of 345 observations generating an alarm in week one, 21/356 observations in week two, and 28/374 observations in week three. *Pregnancy length*'s maximum number of observations generating an alarm was 95/374 in week seven. The indicator *Percentage of dead piglets per farrowing* was the second earliest indicator with more alarms being detected, with 14/345 observations generating an alarm in week one, 15/356 observations in week two, and 18/374 observations in week three. However, this indicator and *Live piglets per farrowing* have much fewer alarms generated then all other indicators. The indicator *Mummified piglets per farrowing* was the one with the highest number of alarms triggered, despite being few at

an early stage. From week four, the number of alarms start to increase considerably with 53 out of 341 observations generating an alarm in week four, 302/342 observations in week five, 339/370 observations in week six, and 357/374 observations in week seven (maximum number of alarms triggered). Lastly, *Piglets weaned per litter* was the indicator which generated fewer alarms in an earlier phase, with only 7/380 observations generating an alarm in week one, 5/373 observations in week two, and 4/379 observations in week three. The maximum number of observations generating an alarm was 86/417 in week twelve.

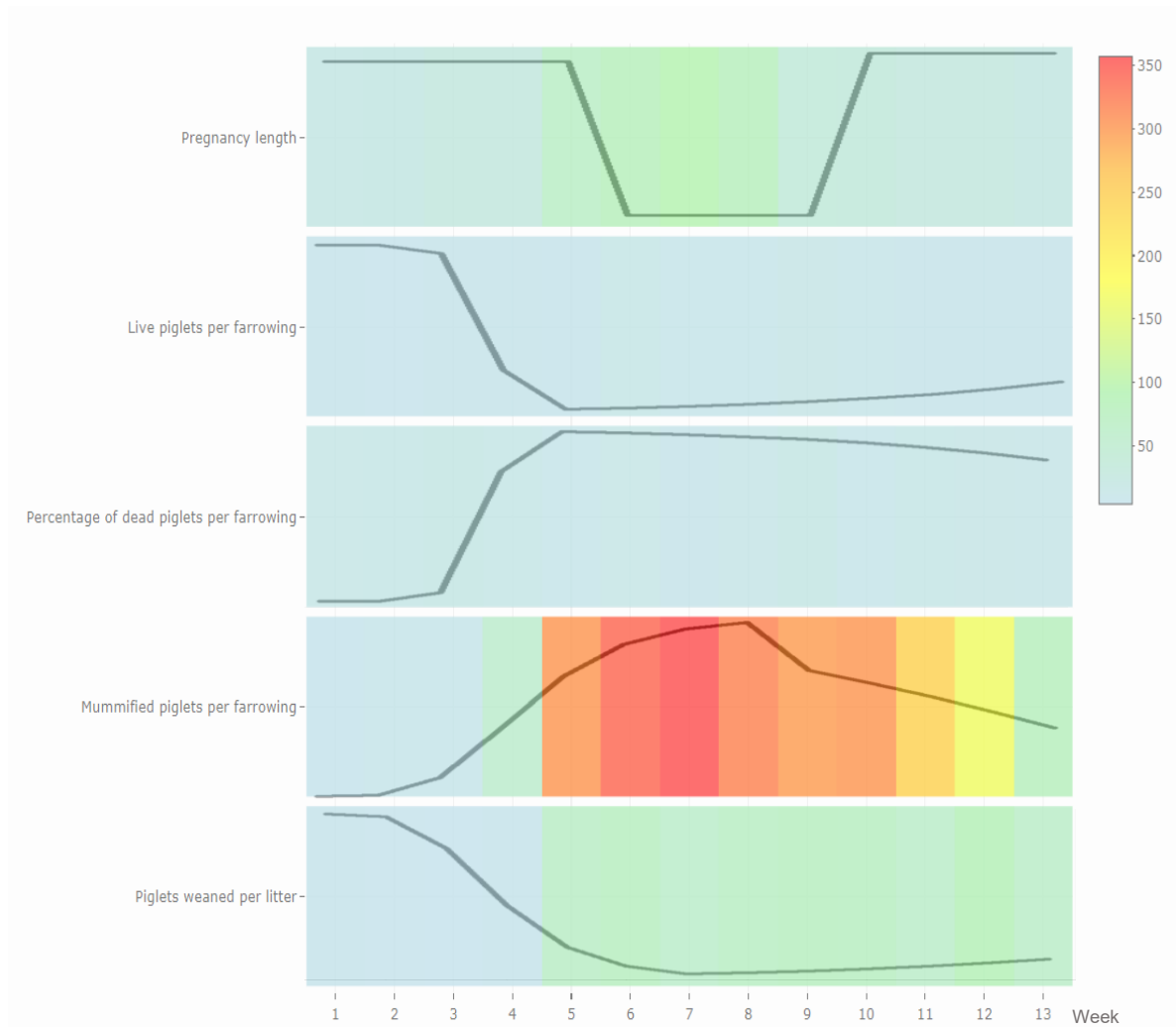


Figure 12: Total number of alarms triggered by the Shewhart control chart in each week of the injected outbreaks, superimposed for all twelve quarters simulated, on the continuous indicators. The corresponding log-normal curves used for aberration injection are also depicted.

6.2. Interactive user interface

The results of the created SyS system applied to the farm under study were joined in a user-friendly interface. In the developed dashboard, all indicators were grouped by the categories above described, and presented on a menu at the dashboard's top. The user can click in one of them and choose if they intend to look at the indicators related to the health or management sections. Within each category, a bar plot for each of the time-series were added. The indicators in which SyS was applied, also the SyS results were plotted. The interface is interactive and the final result is a portable HTML page. Dashboard's layout is shown in Figure 13, with the indicator *Time to reservice* being exhibited.

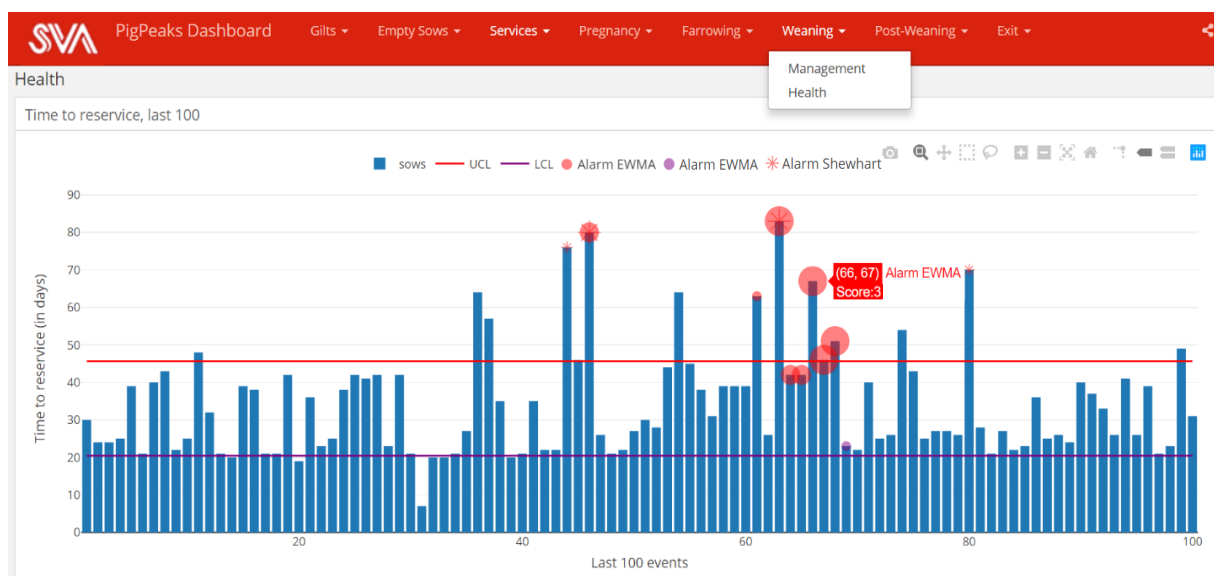


Figure 13: Dashboard.

7. Discussion

Many production problems and infectious diseases present diffuse signs. The assumption behind syndromic surveillance is that these anomalies can be detected when the data are compared to similar data from unaffected populations and the information is monitored continuously (Shmuel and Burkom 2010). In this work, we have developed a SyS system to implement in animal health based on production data. Production data was used since it is considered to be a promising data source, and has not been used extensively in SyS systems in animal health surveillance (Dórea and Vial 2016). We explored production data for swine, voluntarily provided by a Swedish reproduction farm, in order to construct a management monitoring and early disease warning system for the farm under study, that could be also reproducible to other swine reproduction farms.

Production data recorded regularly could be used to construct health and performance indicators. When data was organized into those indicators, it was noticed that we were dealing with two different time-series types. Some indicators were discrete time-series, the time-series typically used in SyS, and some other indicators were better expressed as continuous time-series. Methods for aberration detection have been extensively documented in literature only for the former, and therefore we started by applying the available methods to those indicators that fit the expected format of a discrete time-series.

As this project aimed to detect both early signs of disease introduction and management errors, aberrations important to be detected are expected to occur as spikes or as a slow increase/decrease in the data. Therefore, this project combined two algorithms that had demonstrated solid performance in detecting those shapes (Yahav et al. 2011). The combination of EWMA and Shewhart control charts seems to be a good option since they exhibit complementary performance, aberration shapes missed by one algorithm are generally picked up by the other (Montgomery 2009; Dórea, McEwen, McNab, Revie, et al. 2013). For each algorithm, and for both weekly and continuous time-series on which SyS was applied, three detection limits were used as proposed in Dórea, McEwen, McNab, Sanchez, et al. 2013. Therefore, the outcome was transformed into a scoring system, rather than a binomial signal indicating whether a potential outbreak signal or management error was present or not. This decreases the error caused by setting arbitrary limit values. Moreover, allows the user (farmer, advisor, analyst, veterinarian) to get used to what are the right limits for each indicator, based on a “severity” score. In the future, more indicators can be incorporated into the system without the need to perform major changes to the statistical analyses.

The control charts chosen were trained against a baseline constructed from the historical data, for each indicator individually. The quality of the first 200 weeks of data available was deemed not accurate for the training period, so they were removed. Despite that,

even after deleting the beginning of the time-series, the amount of data left was considered enough for the baseline, and to test the system.

To evaluate the system, first the performance of both control charts in the absence of outbreaks was assessed. In the weekly indicators, *Reservices per week* and *Piglets weaned per week*, both had a 0% false alarm rate with the Shewhart control chart. This should raise a concern about whether the detection is working well on this indicator, or the detection threshold should be lowered. After analysing the detection performance in the presence of outbreaks for these indicators in comparison to the others, it was concluded that Shewhart detection is performing fine in these indicators. On the other hand, the indicator *Abortions per week* had the highest false alarm rate (13.3%) for both control charts. After analyzing this indicator's data, we realized that the farm under study had very few abortions in the historical baseline (Figure 4). So, whenever there is an abortion, the algorithms consider it to be an alarm. While the control charts chosen represent robust coverage of expected outbreak signals for most typical discrete time-series, they are likely not a good fit for this series with very low counts. Therefore, in the future, control charts adapted for rare events should be investigated. As suggested in Mandl, Reis, et al. 2004 a false alarm rate of 3% (97% specificity) was parameterized as the maximum false alarm rate for the system to be usable, because more than that many false alarms are going to be generated, and the users will stop trusting the system. Other than *Abortions per week*, the false alarm rate was only higher than 3% for the combination of *Piglets weaned per week* with EWMA and for *Death in weaning piglets per week* with Shewhart, being the false alarm rate calculated using the lowest detection threshold. These combinations are cases where the scoring system can be used to adjust the expectation of what is a true alarm. The scoring system applied will allow a higher score to be used as a real alarm for these combinations (for instance only consider alarms ≤ -2 or ≥ 2 , respectively), while keeping the same detection thresholds applied to all other weekly indicators in the system. As seen above, in order to reduce the false alarm rate a higher limit has to be applied. However, by setting a higher limit also sensitivity is going to be decreased. Therefore, it is essential to achieve a suitable balance between false alarm rate and sensitivity, even though it is a complex process.

The SyS system created has many more time-series than any other reported in the literature, so there is a risk that every week there will be at least one false alarm. Therefore, in the future, research should focus on how to combine the evidence of multiple indicators to have one overall alarm/assessment of the risk of an ongoing outbreak.

Then, the ability to detect real outbreaks was tested with a simulated PRRS outbreak. For the baseline behavior of the indicators, it was chosen to not use artificial data, but the three years of historical data available for each indicator. This resulted in the possibility to simulate outbreaks against twelve different quarters of basal farm behavior data, giving twelve different scenarios to test the system in terms of noise and data distribution. For the creation of the

hypothetical outbreak signals, rather than simulating defined artificial shapes and applying these same shapes to all indicators, we tried to use evidence from the literature to create outbreak signals that closely represent the expected behavior of each particular indicator under a real scenario of PRRS spread within a farm. As a consequence, when interpreting the results, we must keep in mind that the timeliness and sensitivity observed are a combination of the statistical performance of the control charts for each indicator, as well as the choices made regarding which series contained earlier or stronger outbreak signals.

For weekly indicators, it is visible in Figures 7 and 8 that both algorithms have demonstrated good and similar performance in terms of timeliness when alarms within each indicator are evaluated against their individual outbreak signal, also shown in these figures. However, regarding their sensitivity, Table 6 shows that Shewhart control chart had a relatively good performance for indicators *Abortions per week* and *Deaths in weaning piglets per week*, but for all other indicators it had fewer alarms triggered than EWMA and more quarters without any detection, its performance being much poorer. This was expected since EWMA is supposed to perform better with these slow increasing/decreasing signals.

The weekly indicator with the highest sensitivity was *Piglets weaned per week* with the EWMA control chart, having the highest number of alarms generated and all quarters triggered at least two alarms. It was followed by *Abortions per week* with both algorithms showing equal sensitivity. However, and as addressed before, both indicators had a false alarm rate higher than 3%, meaning that their sensitivities are including false positives. *Reservices per week* with EWMA was the combination with the false alarm rate lower than the recommended, with the highest sensitivity. Regarding EWMA timeliness, the earlier indicator was *Abortions per week*. Nevertheless, since this indicator has a high false alarm rate, the farmers might ignore *Abortions per week*'s alarms. Still, those alarms would be accompanied, since week one, by *Piglets weaned per week*'s alarms which increased every week, and from week three the farmer would also notice alarms of *Deaths in weaning piglets per week*. *Reservices per week* gives additional consecutive alarms but just from week six. Comparative timeliness among indicators was very similar to the Shewhart algorithm, but as its sensitivity was worse, Shewhart's timeliness was not our focus.

The SyS system's results for weekly time-series show that this PRRS outbreak, regardless on which quarter had happened, had a big chance of being noticed, and already on week two, using the indicators chosen. Although not all indicators are equally effective, the combination of the five gives a useful overview of farm animal health regarding this disease. Silva et al. 2017 reported that PRRS early detection was higher for *Abortions per week* than for any other indicator, similar to our results. However, and as addressed before, ours *Abortions per week* results might not be very reliable. On the other hand, it was reported by Engler et al. 2009 that pre-weaning mortality rate, in our study described as *Deaths in weaning*

piglets per week, had detected a change in the process from four weeks to two weeks before PRRS was diagnosed, corroborating our findings of these indicators having good performance for early warning.

In the process of trying to apply SyS to continuous time-series, many difficulties arose. First, how to assess the presence of temporal effects and, when present, remove them, when the observations are not grouped by a specific temporal unit. Second, how to develop the R package {vetsyn} to work for continuous time-series, enabling the definition of what composes the baseline, guard-band, and evaluate windows, and to adopt the prospective SyS framework on them. Also, letting {vetsyn} feature of keeping an outbreak-free baseline available for continuous time-series as well. Third, how to interpret SyS system results when applied to continuous indicators. As opposed to what happens on weekly time-series where there is one observation per week, and consequently, there is or there is not one alarm per week, in continuous time-series observations are not aggregated per time unit. Therefore, and as the number of alarms is intrinsically dependent on the number of observations, it is required to define a threshold that says how many detected alarms in a given time period is considered a real alarm. It was beyond the scope of this work to solve all those issues, and further research will be needed in the future to figure them out. Focus was given in producing the base for this research by applying already available methods to the continuous time-series indicators, and reporting all the challenges we came across and opportunities considered relevant.

Two performance measures used to evaluate the discrete time-series, false alarm rate and timeliness, could be reproduced exactly the same for the continuous indicators. However, we could not calculate sensitivity for these time-series because we would need to define what is an alarm. Therefore, the performance of control charts was assessed only based on false alarm rate and timeliness.

The false alarm rate was the highest for the indicator *Pregnancy length* with EWMA (14.8%). This indicator's observed values had a low range of variation, ranging from 107 to 128 days on the original data (Figure 4). It is likely that this low variability resulted in narrow confidence intervals, and as a consequence the detection statistic can build deviations based on even small noise. Therefore, for this indicator, only alarms of an even greater magnitude than what was used in this study should be considered in the future, or the EWMA parameter (λ) could be changed for a larger value, since larger λ values are recommended for detecting larger shifts. When the Shewhart was applied to this indicator the issue addressed before did not happen. This is due to Shewhart's detecting properties being more efficient at capturing singles spikes in the data. Therefore, the problem of this indicator's values having a low range of variation is not concerning for Shewhart, since alarms were generated just when a significant difference between the baseline and the observed values was noticed. The false alarm rate was also high for *Mummified piglets per farrowing* for both control charts. In the

case of EWMA, with all three detection limits used, the false alarm rate is still going to be bigger than 3%. Again, an even higher detection limit might be used in the future when applying EWMA, and indeed for the Shewhart algorithm considering only alarms of absolute magnitude 2 or more would give an acceptable false alarm rate. More likely, however, the issue should be addressed with another choice of control chart. Similarly to what was reported for the number of abortions per week, this indicator represents a rare event on the farm, and performance of control charts better suited for this type of time-series should be investigated in further steps of this work (but outside the scope of the current investigation). For *Piglets weaned per litter* with EWMA the false alarm rate was high as well. We suspected that it is due to the low range of variation of this indicator (mostly between 10 and 14 piglets weaned per litter) as happened with *Pregnancy length*. Therefore, in the future, lambda could also be changed for a larger value, or only the highest score should be considered as a real alarm for this indicator. Lastly, *Percentage of dead piglets per farrowing* has also a false alarm rate higher than 3% for the EWMA. In the future, real alarms could be considered only when the score is ≥ 2 , while keeping the same detection thresholds applied to all other continuous indicators in the system.

In terms of timeliness, both control charts exhibited again good performance, although the EWMA often notices changes earlier. Despite not being visible in Figures 11 and 12, this conclusion was drawn by analysing the results directly. This was expected since EWMA is known for its performance of detecting gradual changes, meaning that this control chart is supposed to detect aberrations sooner.

Regarding the timeliness of the continuous indicators, the earliest indicator was the *Mummified piglets per farrowing* with the EWMA control chart. But, once again, as this indicator had a high false alarm rate, those alarms had a big chance of being ignored by the farmers. Nevertheless, those alarms would be followed, since week one by *Percentage of dead piglets per farrowing*'s Shewhart alarms. From week three, by *Percentage of dead piglets per farrowing*'s EWMA alarms, and from week four by *Live piglets per farrowing*'s and *Piglets weaned per litter*'s EWMA alarms.

The continuous time-series allow individualized follow-up of each animal and even individual farm events. It was expected that the indicator *Mummified piglets per farrowing* was the one which triggered the most alarms in an earlier phase, since PRRS can manifest as a marked increase in the number of mummified piglets (OIE 2008). Silva et al. 2017 reported that prenatal losses, in our study described as *Percentage of dead piglets per farrowing*, had an alarm signal one week before the detection of PRRS to three weeks after, also corroborating with the present study's results of this indicator being useful for early detection. Therefore, by adding continuous indicators to the system we constructed a better system, with more data, and providing the possibility of using the data differently. This could be useful to construct new

indicators that are not commonly used for analysis, such as the interval between events. In this system assessment, the continuous indicator *Days between farrowings* would be relevant to help the detection of PRRS outbreak, but it could not be used because to the author's knowledge no information about how this indicator would behave during a real PRRS outbreak was available in the literature. Our results showed that including the continuous indicators in the monitoring will add value and timeliness to the system, even if further research is needed to adapt the temporal monitoring methods and, in particular, translate aberration detection results into alarms and interpret their evidence.

Monitoring production data was useful to identify signals associated with PRRS infection, but this tool could also support the identification of non-infectious sources of variation or another infectious disease, since indicators based on production data are not disease-specific. Although this SyS system has not been tested for management errors detection, we believe that the system is able to detect them. Since these management errors are usually registered as spikes in the data, they will be therefore detected by the algorithms chosen, mainly by the Shewhart control chart.

Veterinarians' and animal technicians' timeliness and compliance may be crucial measurements to evaluate the effectiveness of the SyS system (Amezcuca et al. 2010). Therefore, it is important to keep them involved by reducing the time demand of reporting health related data (Dórea et al. 2011). As such, by using production management software WinPig to electronically collect the data, we are working with electronic equipment already present on the farm, which they are already used to, meaning that reporting could be easy and quick. Moreover, the costs of system maintenance are very low because these data are already collected on a regular basis. When the system is implemented, as soon as the data are reported, the software sends them immediately and automatically to the system, increasing the timeliness. Besides, a user-friendly dashboard was created providing valuable information in the future as a result of the implemented SyS system, which increases user's compliance.

This study show that the system created is useful and has potential. However, the fact of using simulated data (and not real data of a PRRS outbreak) and doing the analyses in one single farm are limitations of this study. Future work is required to overcome these limitations in order to improve the system. Implementation in practice and in real-time will further require development of the monitoring system to work prospectively, while all analyses reported here were performed retrospectively.

For this system's usability in real life, it became critical to develop a dashboard where the SyS outputs could be available for user analyses. This was addressed by developing output charts that combine observed data with the detection scores for the two algorithms, plotted over time, provided in a user-friendly interface. The use of algorithms that can identify different outbreak shapes, coupled with a user dashboard, resulted in a robust system. Additional work

is required for the user to be able to fine tune the system to increase performance individually for each indicator. It could be done by adding a feature in the interface provided, where changing the EWMA parameter (λ) and replacing the detection limits chosen would be possible, allowing flexible customization of alarms. For instance, the reporting threshold for an individual indicator can be increased when several false alarms are being noticed, increasing specificity. Otherwise, the threshold can be lowered to improve sensitivity if a specific indicator requires more intensive monitoring.

Another opportunity in SyS not addressed in this study is to carry out a multivariate analysis taking sows' parity (number of times that have given birth) and seasonality into account. These will be evaluated in future steps of this work.

Other future perspectives of this work are testing the SyS system created with other swine reproductive farms and implement the system in the real world. Therefore, when the system is implemented and once an alarm has been raised, a specialist should be called in to decide whether a true problem exists. If an outbreak is discovered, the veterinarian will be able to investigate the occurrence in order to find the cause, control the spread of the disease and treat the infected population.

8. Conclusion

A digital system to monitor productivity indicators continuously, and alert for changes in reproductive performance at the herd level was constructed for later implementation on a Swedish swine farm. It was demonstrated that it can provide early detection of unexpected trends, and provide a decision tool to support farmers who can detect those signs themselves, allowing them to act timely. Control is therefore triggered before significant economic losses and disease spread occur. All the steps were explained to be readily reproduced, and all the challenges we came across and opportunities considered important to be improved were reported, in order to produce the base for this research. It is hoped this work will further encourage and facilitate the development of other syndromic surveillance systems using production data.

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10. Appendix

Porcine reproductive and respiratory syndrome (PRRS), also named blue ear disease, was first reported in North America in the mid to late 1980s and spread rapidly worldwide. The causal agent is an RNA virus classified as a member of the genus *Arterivirus* (OIE 2008).

As the name suggests there are two syndromes associated with PRRS; reproductive failure in sows, and respiratory disease in young pigs. The reproductive failure is characterised by infertility, premature farrowings, a marked increase in abortions, stillbirths, mummified piglets or the birth of weak piglets that often die soon after birth from respiratory disease and secondary infections. The stage of gestation, the immune status of the sow, and the virulence of the virus determine the effects of infection. Regarding the respiratory disease, the virus has a tropism for macrophages (white blood cells) in the lungs where it reproduces diminishing the immune response. The respiratory syndrome is recognised by laboured breathing, fever, loss of appetite, red discoloration of the body, and blue ears. Younger pigs are more affected than older animals (death can occur in up to 10% or more of sows). There is often high mortality, though infected animals can also show no disease at all. And in spite of the name, the blue ears, due to restricted blood supply, are only a transient sign of the disease (OIE 2008).

PRRS virus (PRRSV) is easily spread following direct contact and virus can be detected in saliva, urine, milk, colostrum, and faeces of infected animals. The disease spreads quickly through a herd over 7–10 days. Transmission by semen can also occur. Pigs born to infected sows may not show disease signs but can still be virus shedders. The virus can also be spread by vehicles or fomites and even insects have been found as a potential source of spread. As well there is the possibility of windborne spread (OIE 2008).

The diagnostic can be confirmed through Virus isolation, RT-PCR (Reverse transcription polymerase chain reaction) testing, and Serological tests. A vaccine exists and is effective. Commercial vaccines, both modified live and killed, have been used and they have been effective in controlling outbreaks and preventing economic losses (OIE 2008).